WEEK 4 · LECTURE

Large language models as research instruments

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The machine, running

LLAMA3.1 · 8B · 32 LAYERS · 8 BILLION WEIGHTS · RUNNING ON THIS LAPTOP



The restaurant was packed on a Friday night, and the food was

A language model does one thing. Given the text so far, it produces a **probability for every token** in its vocabulary — 128,256 of them. The prompt is at the bottom. Click to run it.

1 / 7

The machine, running

LLAMA3.1 · 8B · 32 LAYERS · 8 BILLION WEIGHTS · RUNNING ON THIS LAPTOP



The restaurant was packed on a Friday night, and the food was **delicious**

NEXT TOKEN · THE TEN MOST PROBABLE OF 128,256

_delicious	9%
_excellent	7%
_good	5%
_great	4%
_amazing	4%
_served	4%
_a	3%
_just	3%
_slow	2%
_still	2%

The ten most probable next tokens, of 128,256. **“delicious”** won with 9%; “excellent” and “good” were close. Nothing is written yet: this is a table of numbers.

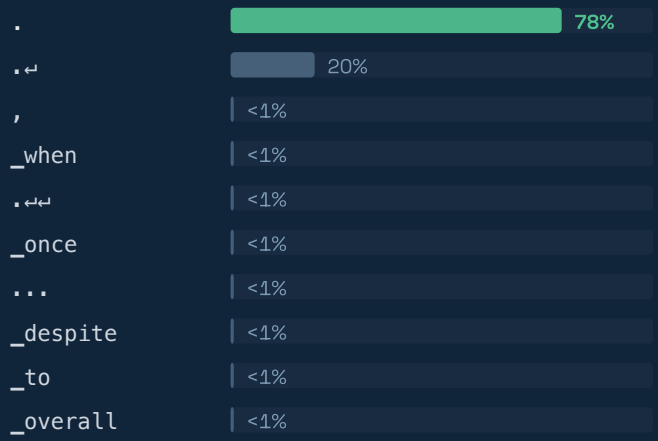
The machine, running

LLAMA3.1 · 8B · 32 LAYERS · 8 BILLION WEIGHTS · RUNNING ON THIS LAPTOP



The restaurant was packed on a Friday night, and the food was delicious. The service was a bit slow, but the staff was friendly and attentive.

NEXT TOKEN · THE TEN MOST PROBABLE OF 128,256



17 tokens, 17 distributions. **Nothing was "written"** : at temperature 0 the most probable token is taken every time, so the same prompt gives the same sentence. Above 0 the model samples from the bars, and the sentence changes from run to run.

The machine, running

LLAMA3.1 · 8B · 32 LAYERS · 8 BILLION WEIGHTS · RUNNING ON THIS LAPTOP



Classify this restaurant review as positive or negative. Answer with one word. Review:
Not good at all.

NEXT TOKEN · THE TEN MOST PROBABLE OF 128,256

Negative	100%
negative	<1%
_Negative	<1%
Neg	<1%
Positive	<1%
N	<1%
_negative	<1%
Neutral	<1%
NEG	<1%
Bad	<1%

A classification is the same operation. **The prompt ends, and the first token of the answer is the label.** Nothing was trained on reviews: the instruction is the codebook, written in words.

The machine, running

LLAMA3.1 · 8B · 32 LAYERS · 8 BILLION WEIGHTS · RUNNING ON THIS LAPTOP



Classify this restaurant review as positive or negative. Answer with one word. Review:
Slow, but worth it.

NEXT TOKEN · THE TEN MOST PROBABLE OF 128,256

Positive	94%
Neutral	5%
Mixed	1%
Negative	<1%
Amb	<1%
Pos	<1%
Ne	<1%
positive	<1%
Neither	<1%
M	<1%

A borderline sentence. The label is still the top of a distribution: **positive 94%** , neutral 5%. The rest of the distribution never reaches your data frame — unless you ask for it.

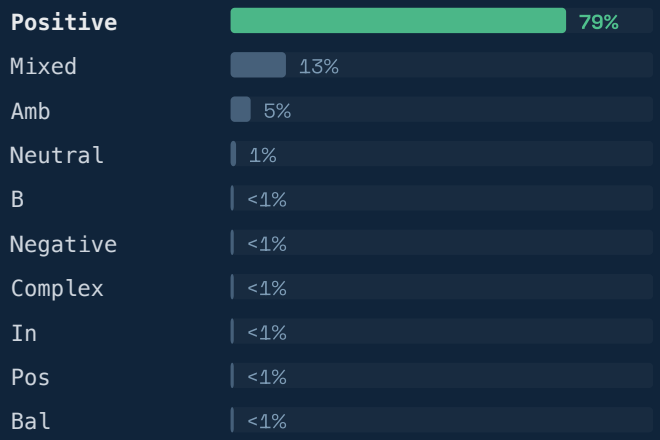
The machine, running

LLAMA3.1 · 8B · 32 LAYERS · 8 BILLION WEIGHTS · RUNNING ON THIS LAPTOP



Classify this restaurant review as positive or negative. Answer with one word. Review: **Flawed and completely wonderful.**

NEXT TOKEN · THE TEN MOST PROBABLE OF 128,256



Here the model is unsure: **positive 79%** , mixed 13% — a label the codebook did not offer. **What you report is one word. What the model computed is this chart.** This lecture is about the distance between the two.

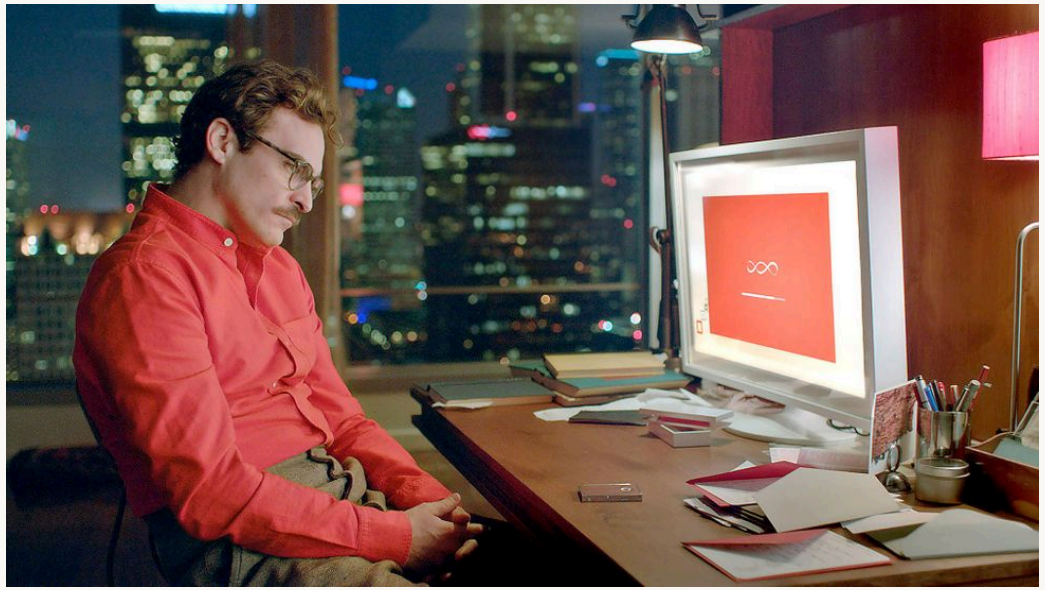
7 / 7

> We may hope that machines will eventually compete with men in all purely intellectual fields. But which are the best ones to start with? [...] It can also be maintained that it is best to provide the machine with the best sense organs that money can buy, and then teach it to understand and speak English. [...] We can only see a short distance ahead, but we can see plenty there that needs to be done.
>
> Alan Turing, 1950

> **Seventy-six years later the machine speaks English. This lecture is about what it is doing when it does, and what that is worth to a content analysis.**

Turing, A. M. (1950). Computing machinery and intelligence. *Mind, 59*(236), 433–460. https://doi.org/10.1093/mind/LIX.236.433 — closing paragraph.

What if you could just ask?



Her (Spike Jonze, 2013). Theodore talks to an operating system that reads his mail, sorts it, and answers back.

- Three weeks of wrangling, word lists, splits, recipes and grid searches — for one label per text
- In 2013 this was science fiction: a machine you describe the task to, in your own words, and it does it
- Since late 2022 it is a product, and since 2024 it runs on the laptop
- The question for a researcher is not whether it works. It is **how can we know** that it does what it is supposed to do?

> **This week the instruction replaces the training. The validation does not.**

Jonze, S. (Director). (2013). *Her* [Film]. Annapurna Pictures. Still reproduced for teaching.

… and it does work

```
library(tidyllm)

abstract <- sci_test$text[1]      # one of last week's held-out abstracts

llm_message(str_c(
  "Which discipline is this abstract from: physics, computer science ",
  "or statistics? Answer with the field only.\n\n", abstract)) |>
  chat(ollama(.model = "llama3.1:8b", .temperature = 0)) |>
  get_reply()
```

- No recipe, no training set, no fitted object: a question in English and a model that runs locally
- The answer arrives in about two seconds

THE ABSTRACT

```
The Reduced PC-Algorithm: Improved Causal Structure
Learning in Large Random Networks We consider the
task of estimating a high-dimensional directed
acyclic graph, given observations from a linear
structural equation model with arbitrary noise
distribution....
```

OUTPUT

```
Statistics
```

```
human label: statistics
```

> Right, on the first try. Whether it is right on the next thousand is Part V.

Last week ended here

- A classifier learns its weights from **labelled texts**; the gold standard we coded is what it learns from and is scored on
- Changing what the model *sees* moved the score more than changing the algorithm: a bag of words at .75, a sentence embedding at .80
- The sentence-embedding model was trained by somebody else, on text we never saw, to keep what a sentence means — and it beat everything trained on our own data alone
- Van Atteveldt et al. (2021): classifiers beat every dictionary and reach no human. The table had one row missing

> **This week the model that read the sentence also writes the label. The gold standard does not change.**

This lecture

I

What changed since 2013

Three advances, the timeline's last screen, the pipeline with one box fewer

II

Inside a transformer

The architecture built in eight clicks; the two Harrys, measured

III

The models

What an LLM is, who makes them, how big, and where your text goes

IV

LLMs as instruments

The prompt as codebook; structured output; what moves the score

V

Four generations, one gold set

Dictionary, classifier, embeddings and language models on the same texts

VI

What to do with it

Images, rules, bias, and the last homework

I · What changed since 2013

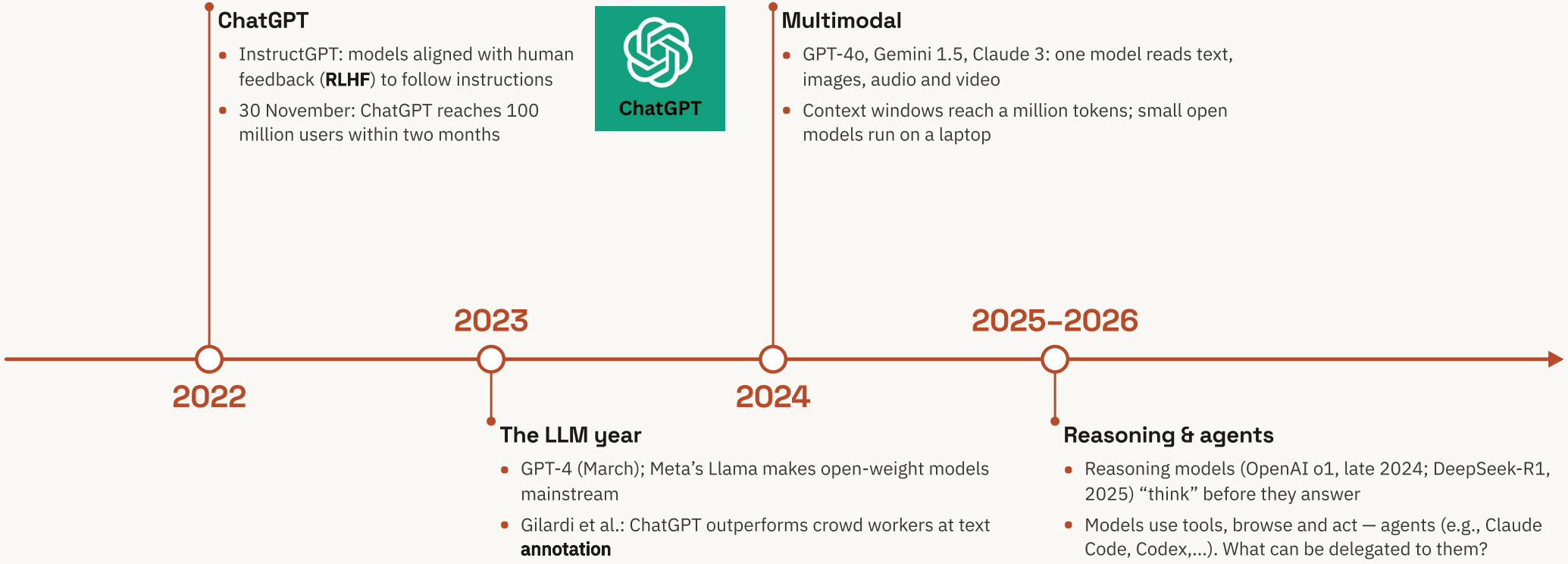
Three advances

- **Representation.** From counts to word vectors to *contextual* vectors: a word's numbers now depend on the sentence it is in
- **Pre-training and transfer.** A model trained once, on an enormous corpus, for a general task — then reused for our task, with a little fine-tuning or none at all
- **Scale.** The same architecture, thousands of times larger, trained on most of the written web; abilities nobody trained for appeared along the way

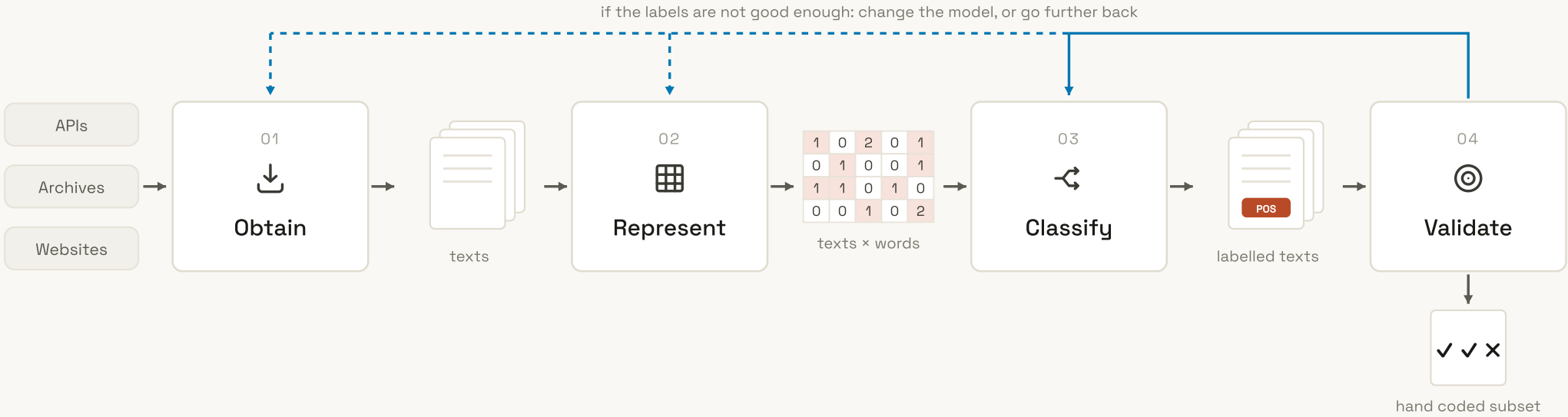
> **The three compound. The instrument we get is a model that reads a text and can be told, in words, what to do with it.**

Kroon, A., Welbers, K., Trilling, D., & van Atteveldt, W. (2024). Advancing automated content analysis for a new era of media effects research: The key role of transfer learning. *Communication Methods and Measures, 18*(2), 142–162. https://doi.org/10.1080/19312458.2023.2261372

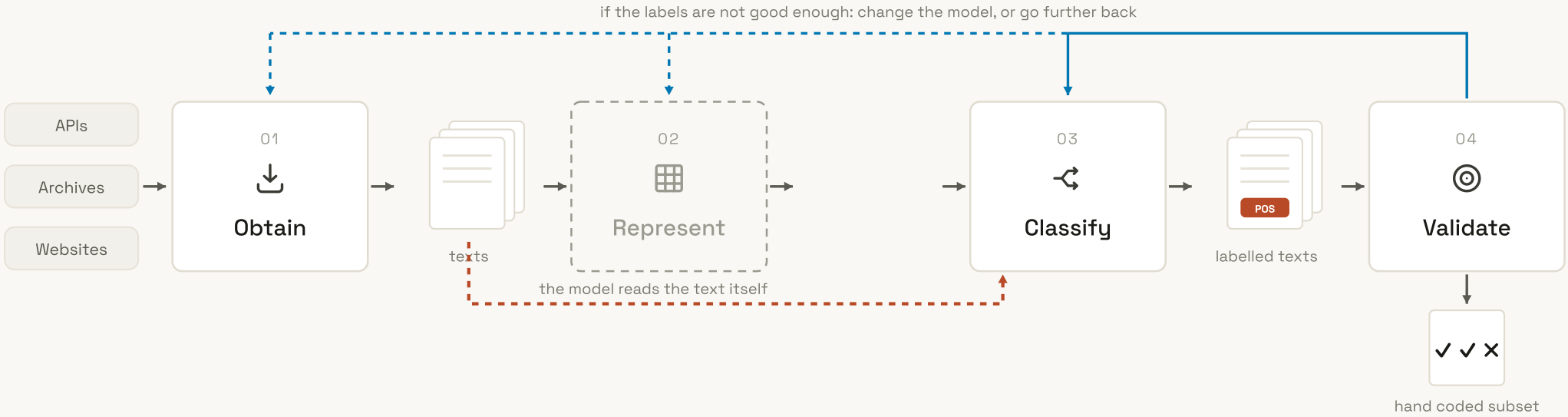
The last screen of the timeline



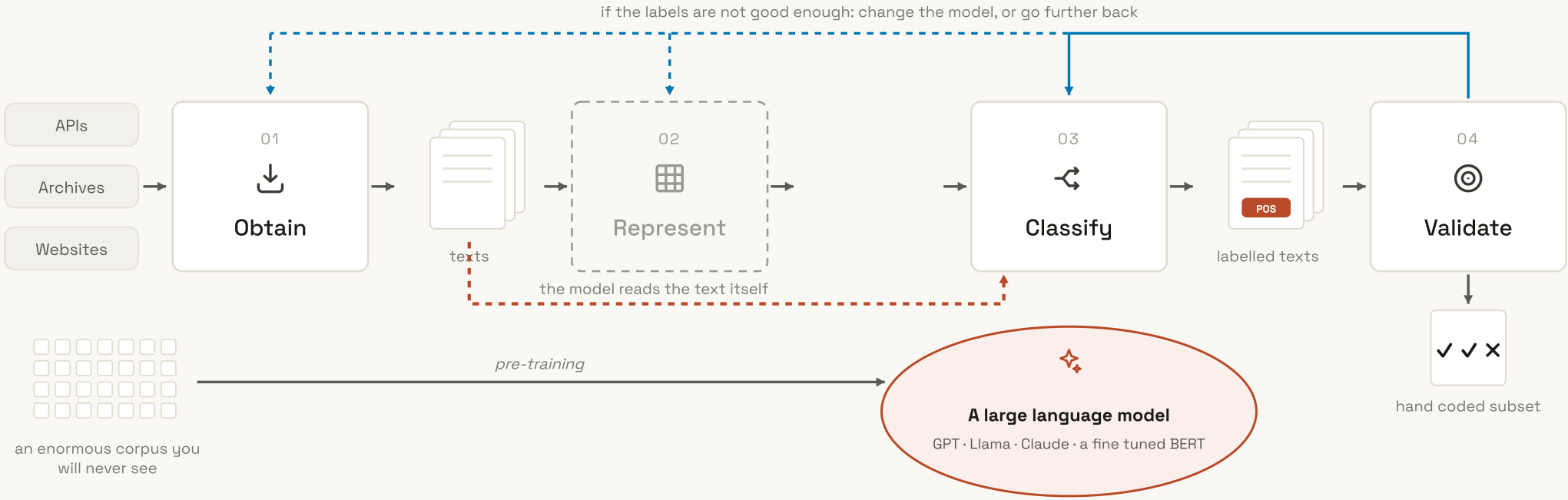
The pipeline with a language model



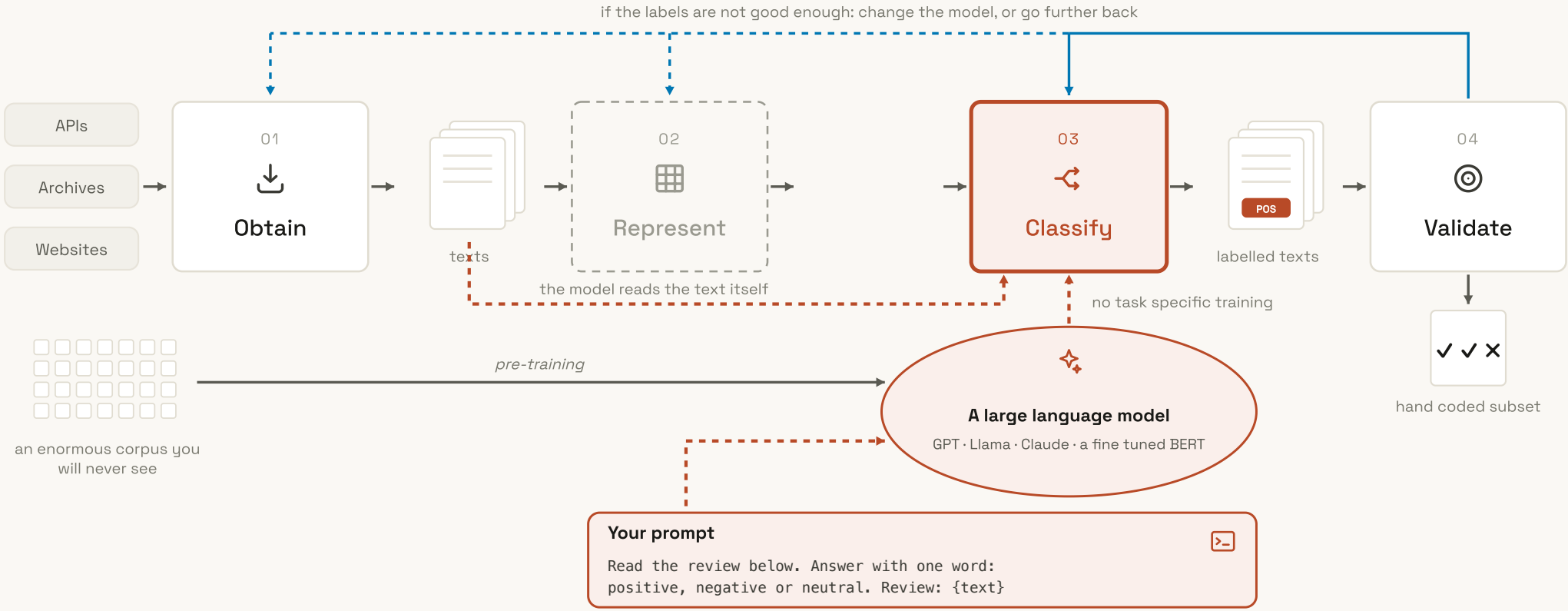
The pipeline with a language model



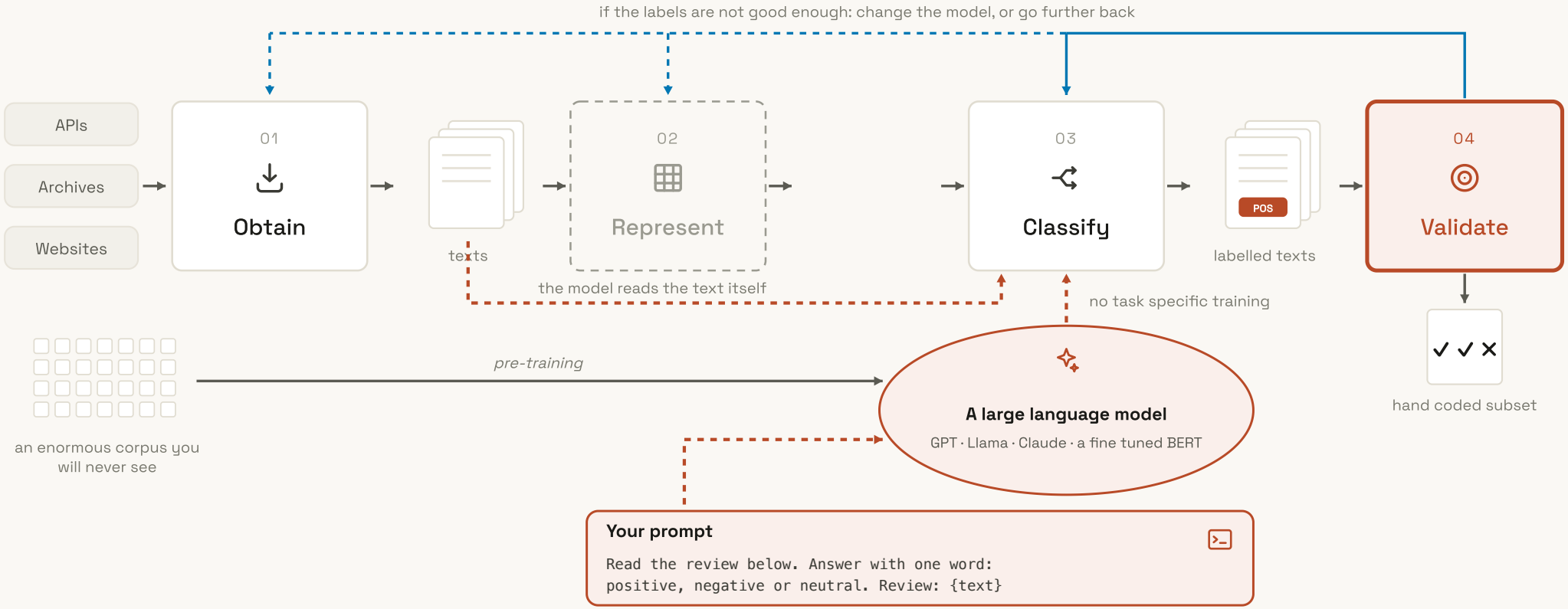
The pipeline with a language model



The pipeline with a language model



The pipeline with a language model



5 / 5

II · Inside a transformer

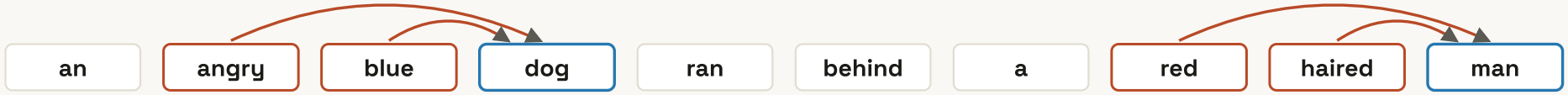
Attention is all you need

- Until 2017 the models that read sequences did so **one word at a time**, carrying a memory forward — slow to train, and forgetful over long distances
- Vaswani et al. (2017), at Google, dropped the recurrence: every word looks at every other word at once, through a mechanism they called **attention**
- The consequences: training could be parallelised, so models could be made much larger; and distant words could influence each other directly
- The paper was about translation. The architecture — the **transformer** — is the backbone of BERT, GPT, Llama, Claude, Gemini and everything else in this deck

> **One architecture, 2017, cited a hundred and fifty thousand times. Everything since is that block, stacked and scaled.**

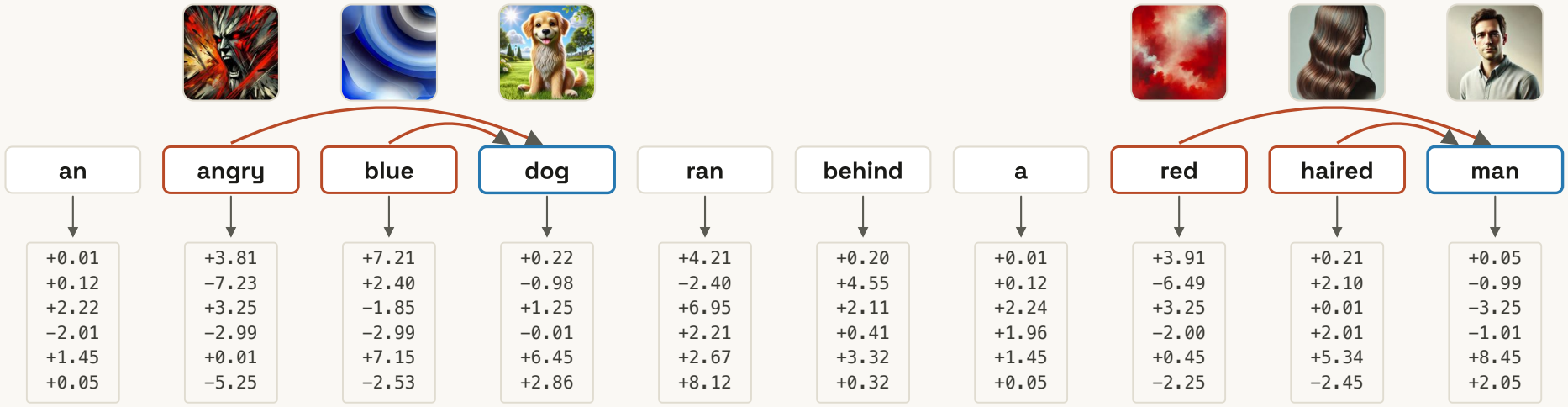
Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems, 30*, 5998–6008.

How a word's vector picks up its context



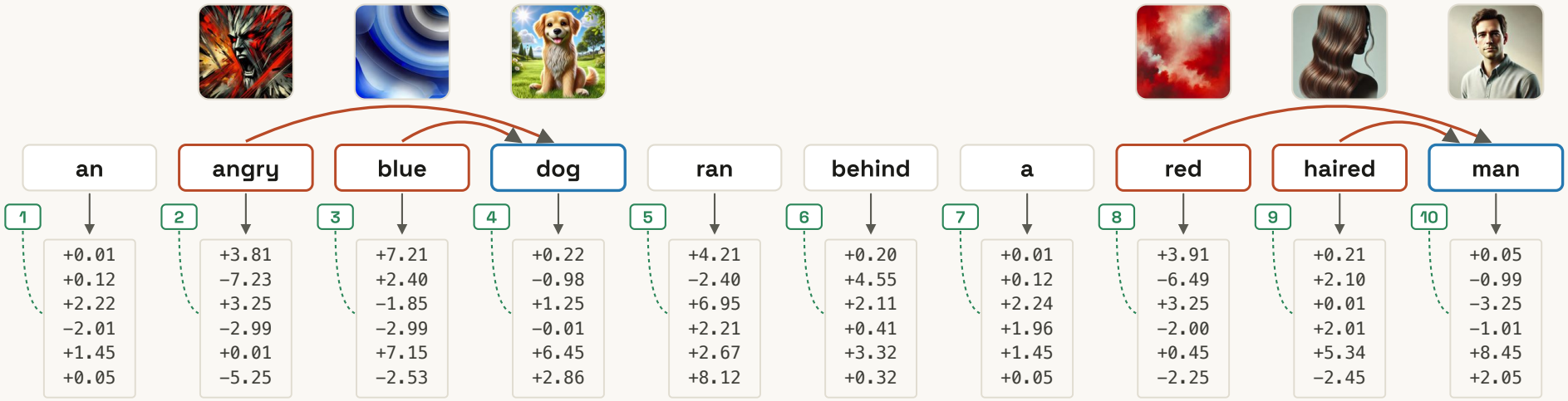
1 / 8

How a word's vector picks up its context

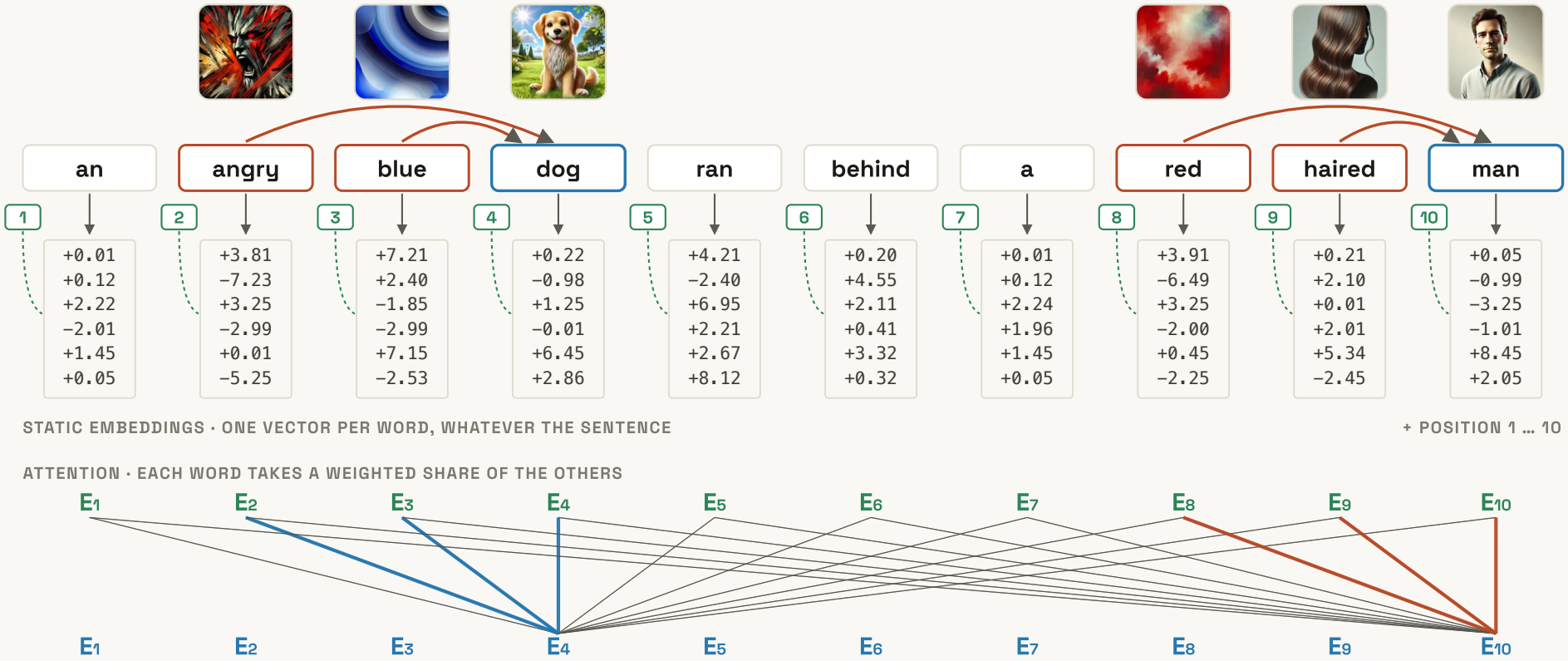


STATIC EMBEDDINGS · ONE VECTOR PER WORD, WHATEVER THE SENTENCE

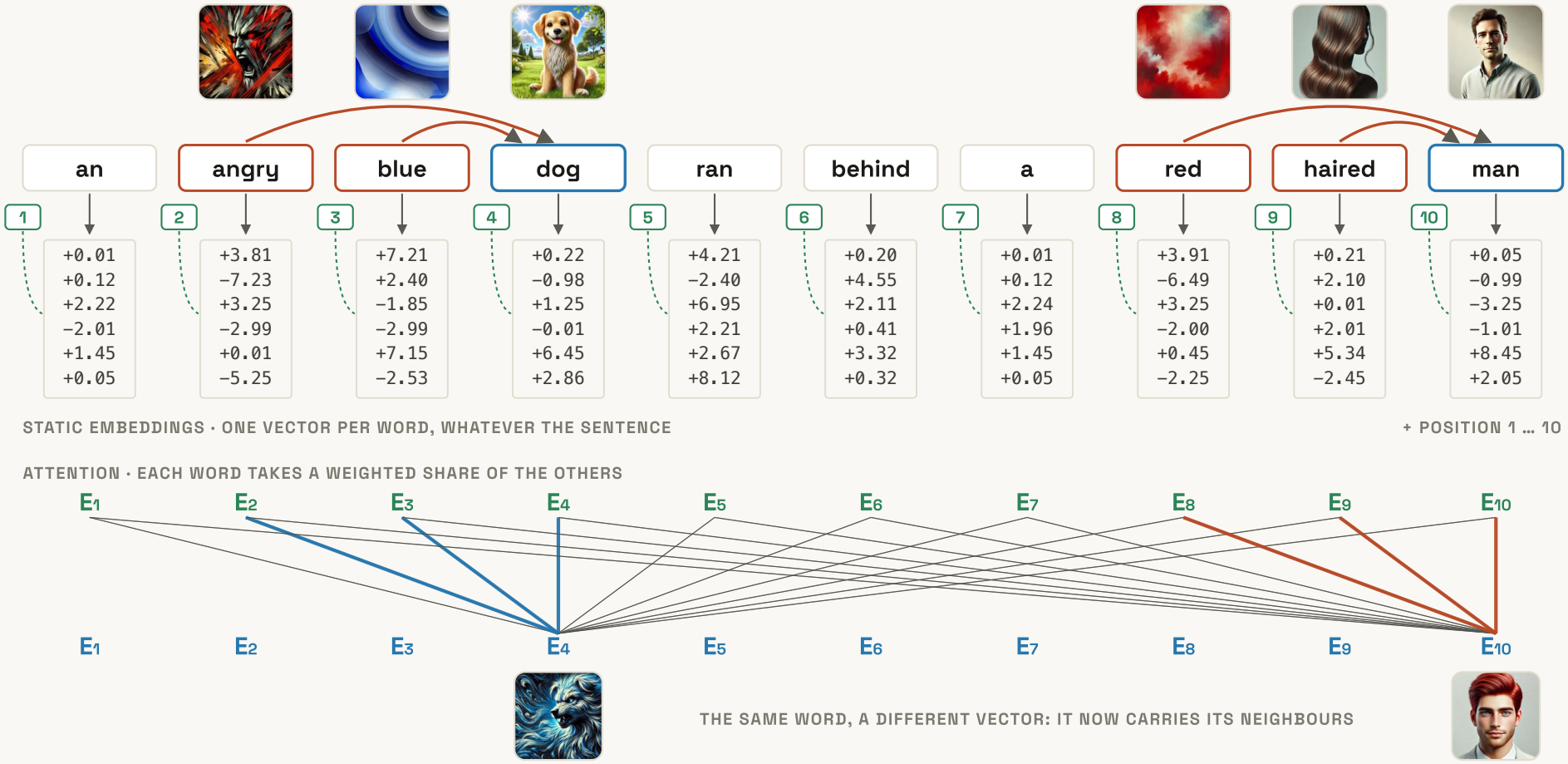
How a word's vector picks up its context



How a word's vector picks up its context

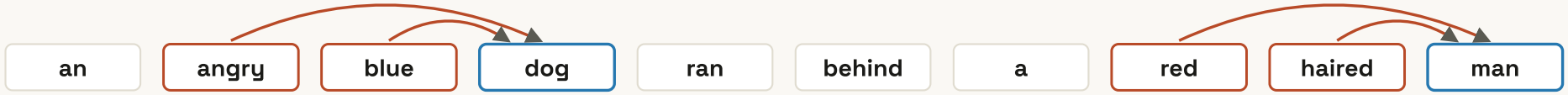


How a word's vector picks up its context



5 / 8

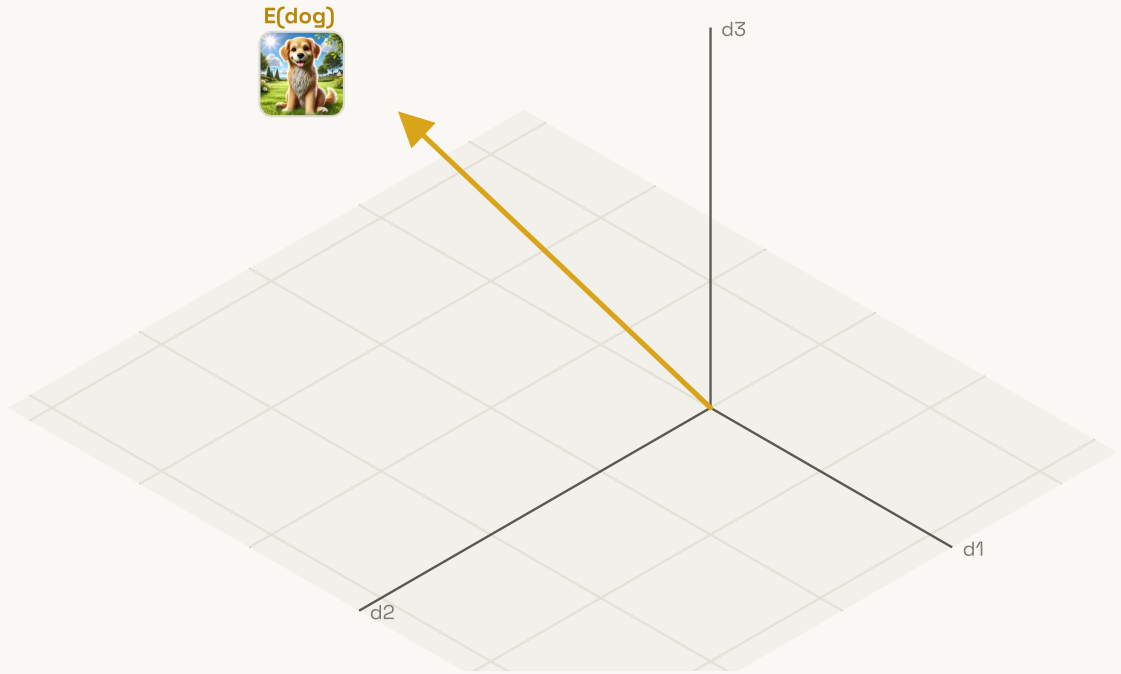
How a word's vector picks up its context



THE SAME VECTOR, MOVING AS ITS CONTEXT IS FOLDED IN

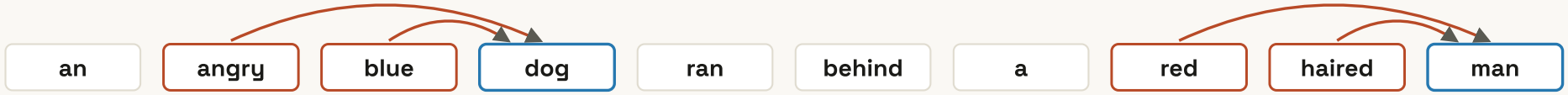
three of the hundreds of dimensions, illustrative

dog



6 / 8

How a word's vector picks up its context

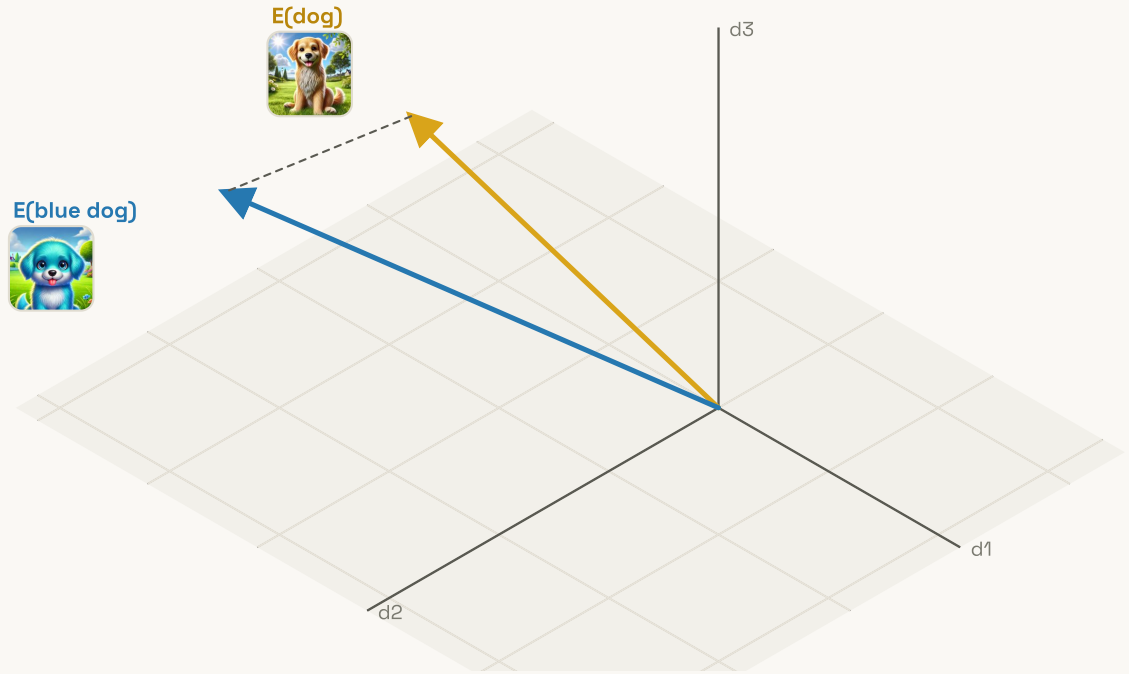


THE SAME VECTOR, MOVING AS ITS CONTEXT IS FOLDED IN

three of the hundreds of dimensions, illustrative

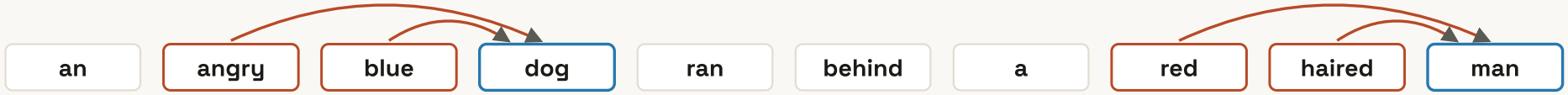
dog

blue dog



7 / 8

How a word's vector picks up its context



THE SAME VECTOR, MOVING AS ITS CONTEXT IS FOLDED IN

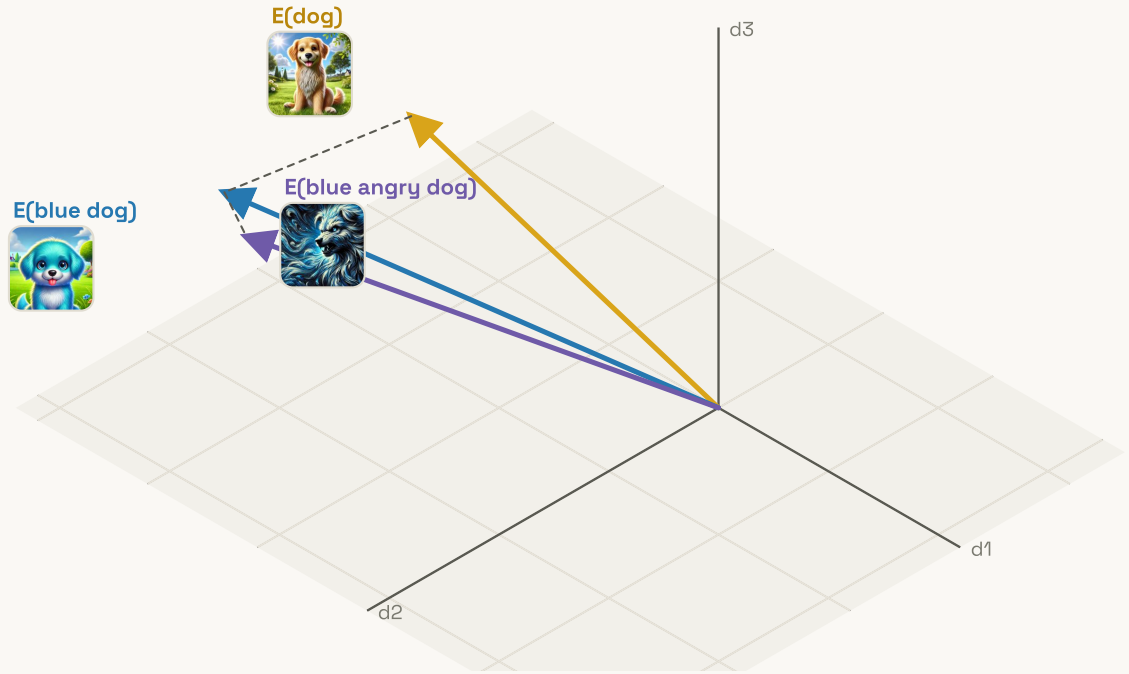
three of the hundreds of dimensions, illustrative

dog

blue dog

blue angry dog

each layer of attention moves it again



8 / 8

Inside a transformer

01 · TOKENS

The sentence is cut into pieces the model has a number for. Nine here; a rare word can be several.

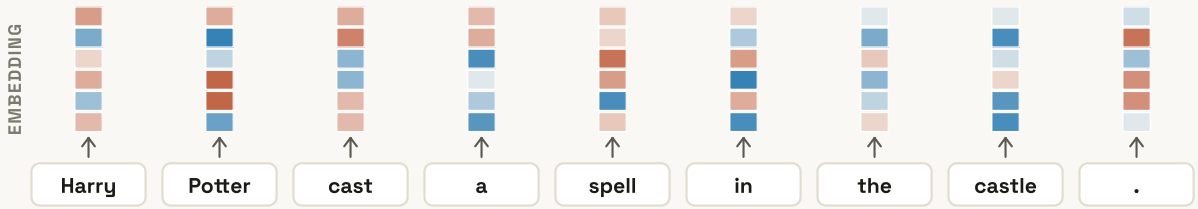
Harry	Potter	cast	a	spell	in	the	castle	.

1 / 8

Inside a transformer

02 · EMBEDDINGS

Each token becomes a vector, looked up in a table learned in pre-training. **Harry's is the same vector whoever Harry is.**

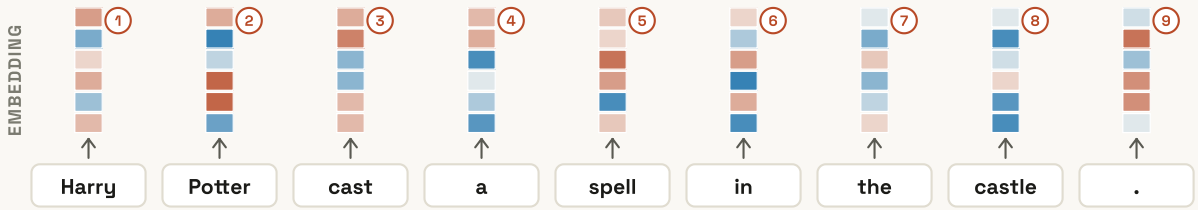


2 / 8

Inside a transformer

03 · POSITION

A second vector says where the token stands and is added in. Without it the model would see a bag of words.

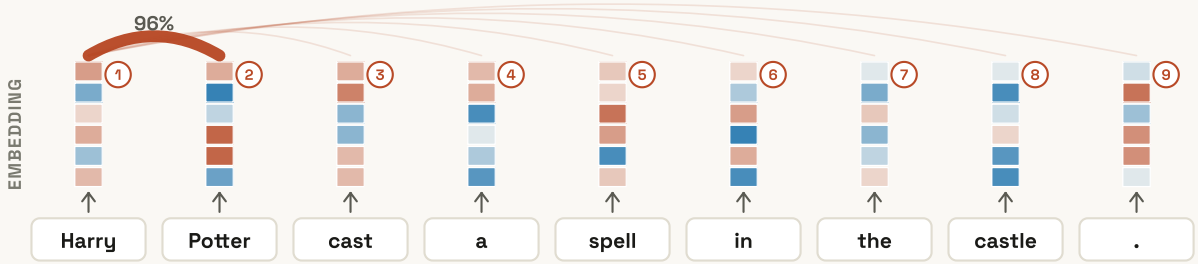


3 / 8

Inside a transformer

04 · ATTENTION

Harry asks every token how relevant it is: a **query** matched against every **key**. BERT, layer 4, head 0, measured: Potter, 96%.

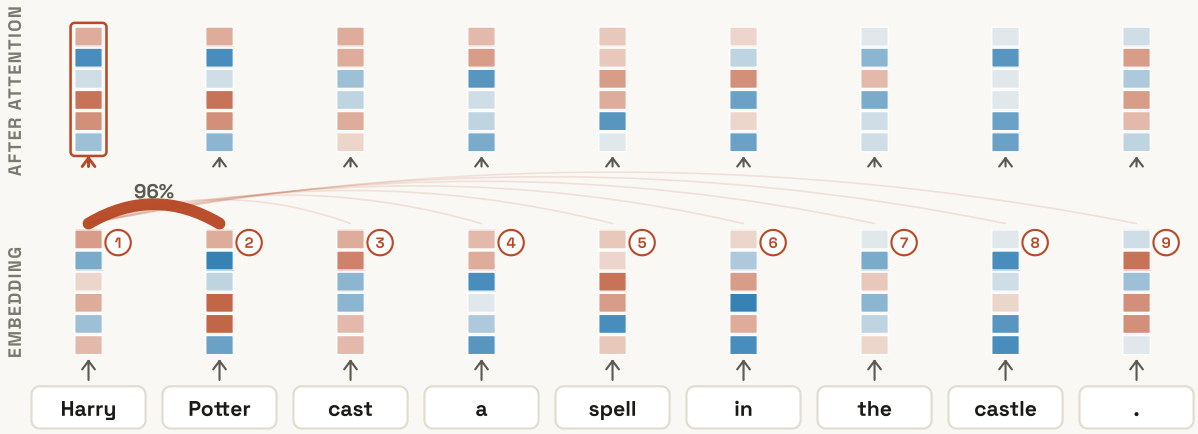


the rest under 1% each · 2.7% parked on a bookkeeping token

Inside a transformer

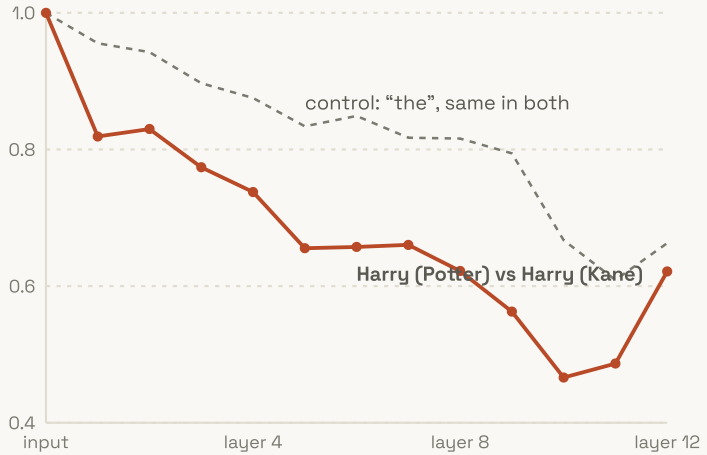
05 · UPDATE

Harry takes a weighted share of what the others carry (their **value**) and its vector changes. **It now contains Potter.** Right: the two Harrys drift apart, layer by layer.



the rest under 1% each · 2.7% parked on a bookkeeping token

THE TWO HARRYS, LAYER BY LAYER · MEASURED IN BERT



similarity of the same word’s vector in the two sentences

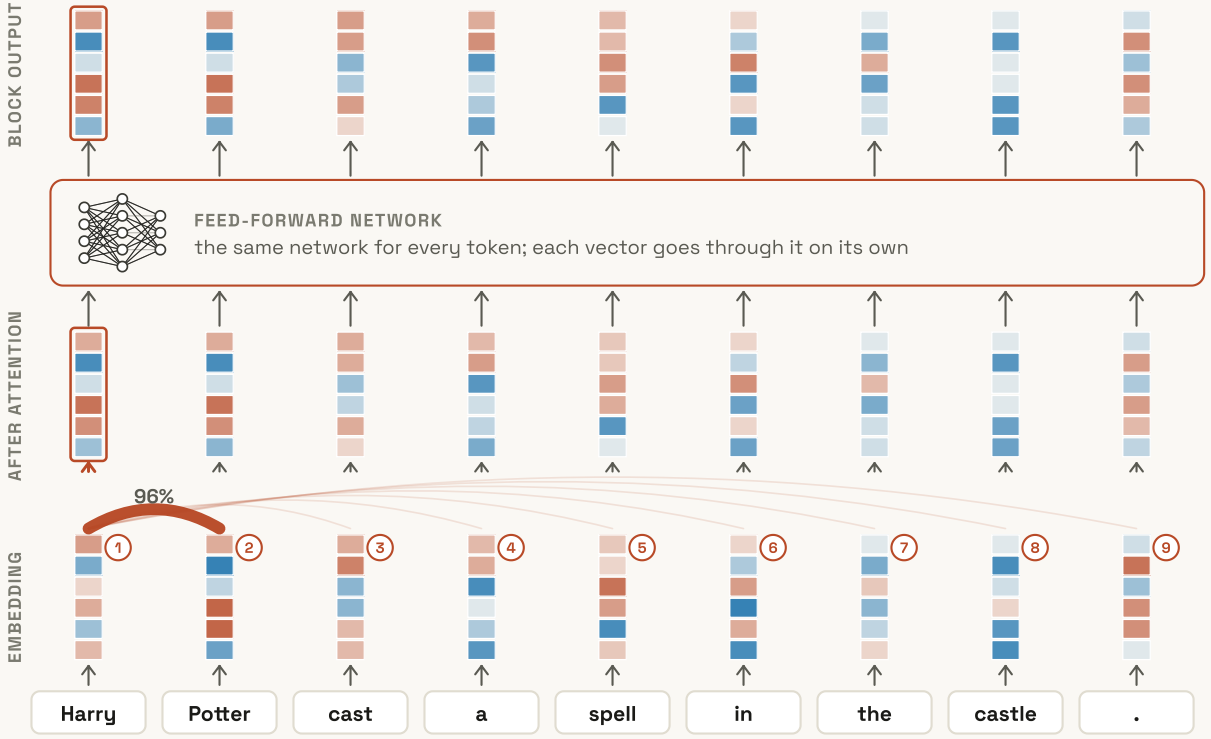
1.00 at the input: one table, one vector. .47 after layer 11: two different Harrys. The vector was rewritten by what stood next to it.

5 / 8

Inside a transformer

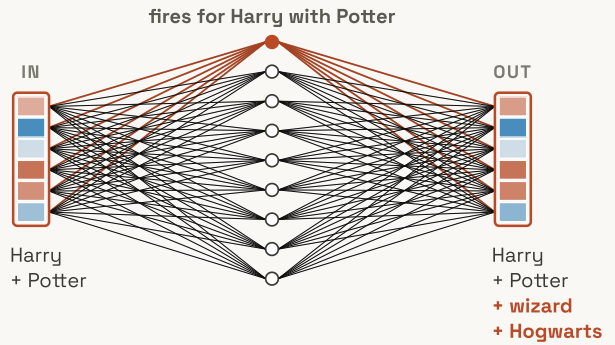
06 · FEED-FORWARD

Each vector then goes through a network like week 3's, **on its own**. Its hidden units respond to patterns and add what the model learned in pre-training: Harry with Potter picks up wizard.



the rest under 1% each · 2.7% parked on a bookkeeping token

HARRY'S VECTOR, ON ITS OWN · ILLUSTRATIVE



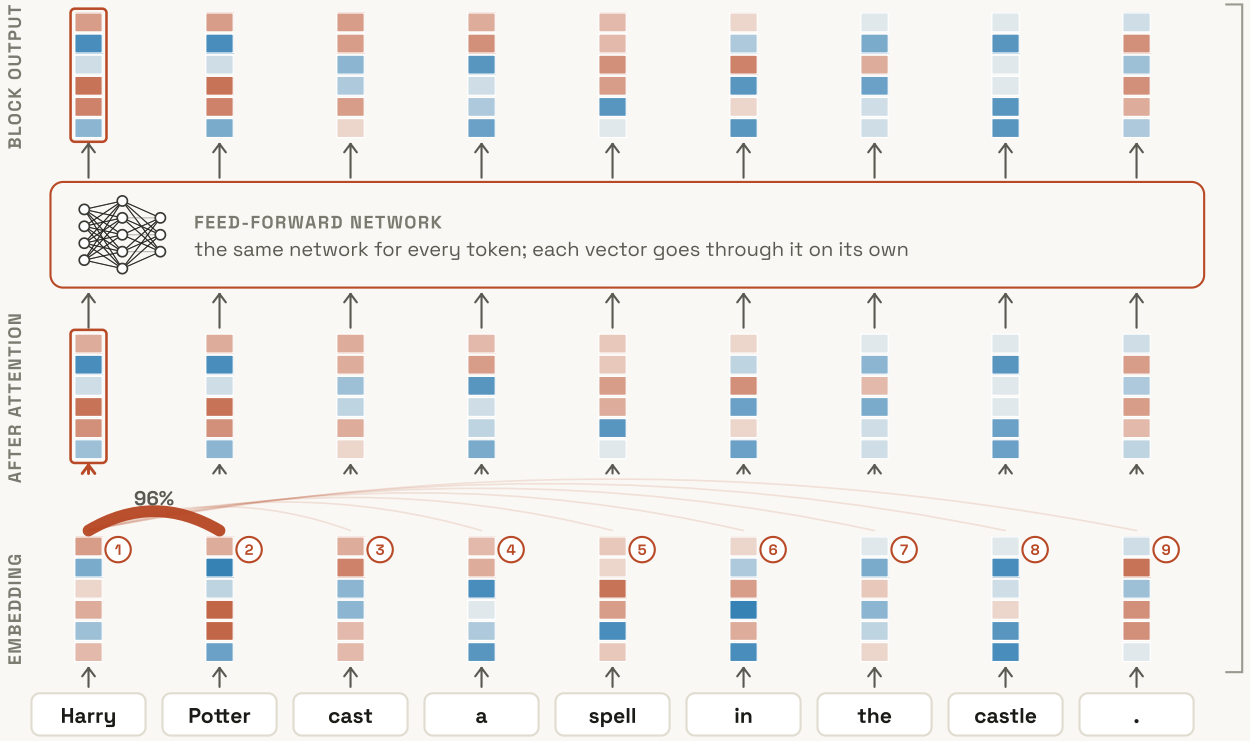
The lit unit responds to Harry together with Potter. The network's output is **added** to the vector: Harry and Potter stay, wizard and Hogwarts join them.

BERT-base has 3,072 such units in every block, Llama 3.1 8B has 14,336.

Inside a transformer

07 · MANY LAYERS

Attention and feed-forward make one block. **Each block works on what the blocks below added**: once Harry carries wizard, the castle can look at Harry and become Hogwarts.



the rest under 1% each · 2.7% parked on a bookkeeping token

THE STACK

attention + feed-forward
attention + feed-forward
attention + feed-forward
attention + feed-forward
attention + feed-forward
attention + feed-forward
attention + feed-forward
attention + feed-forward
attention + feed-forward
attention + feed-forward

WHAT THE BLOCKS DO, ROUGHLY

Upper blocks: what the sentence says, and what may come next.

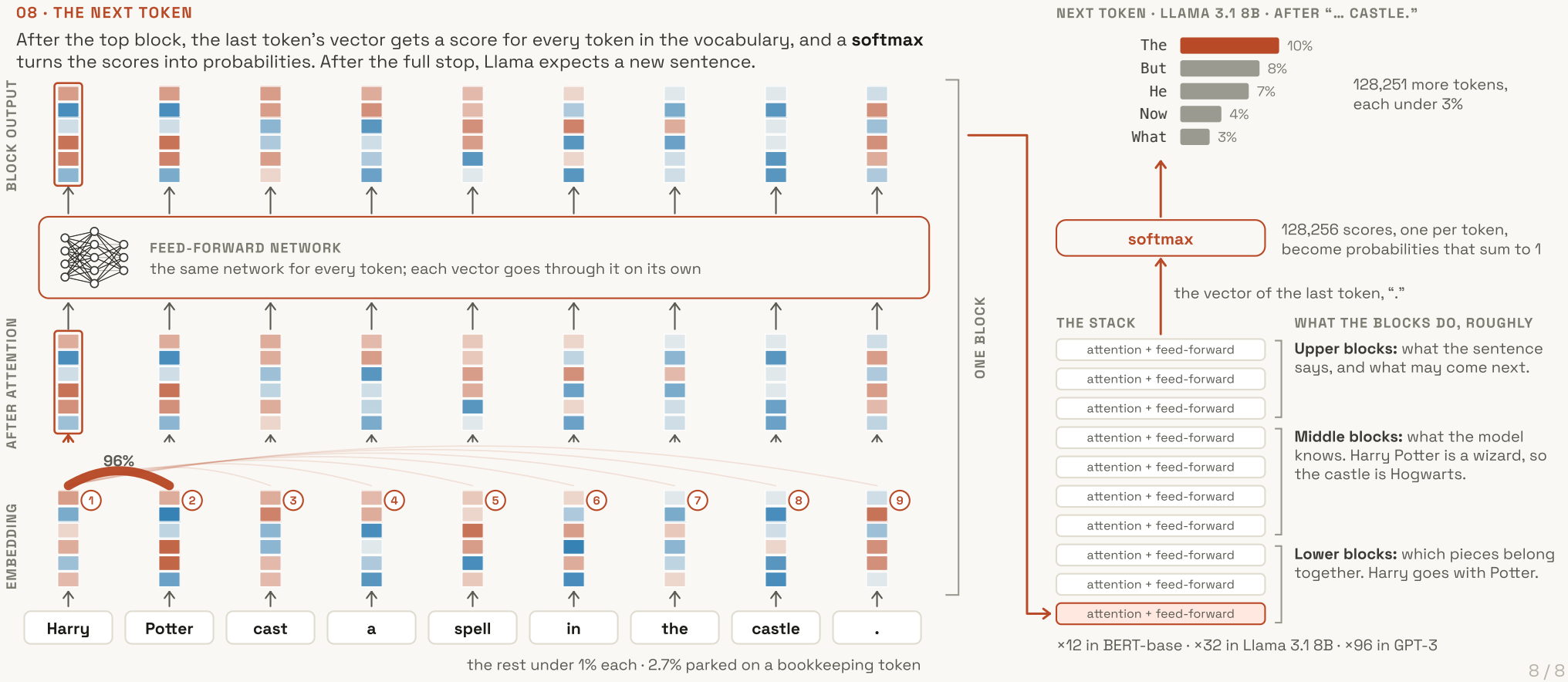
Middle blocks: what the model knows. Harry Potter is a wizard, so the castle is Hogwarts.

Lower blocks: which pieces belong together. Harry goes with Potter.

×12 in BERT-base · ×32 in Llama 3.1 8B · ×96 in GPT-3

7 / 8

Inside a transformer



Which Harry? Measured

DATA · BERT-BASE-UNCASED · TWO SENTENCE PAIRS

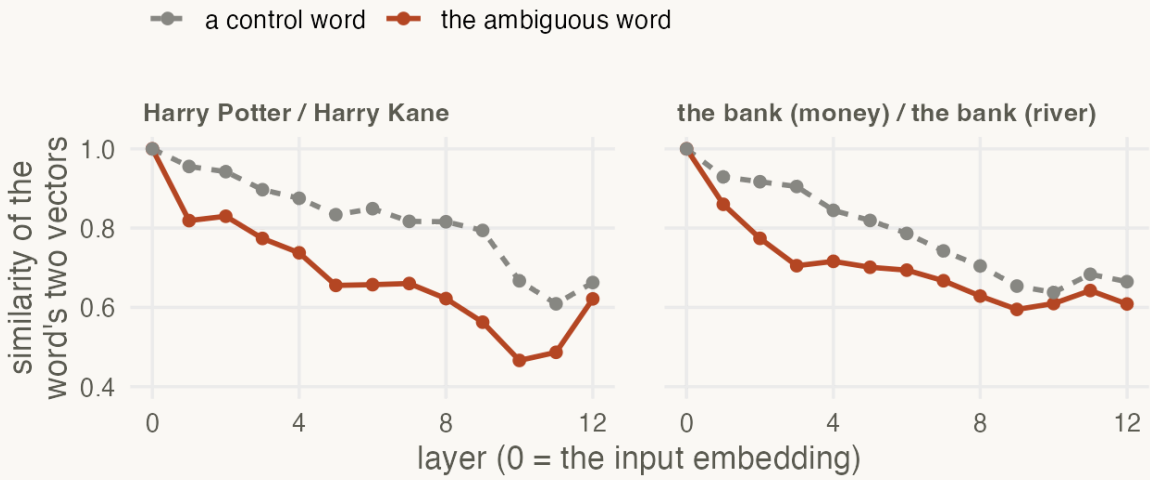
ONE NAME, TWO PEOPLE

Harry Potter cast a spell in the castle.
Harry Kane scored a goal in the stadium.

ONE WORD, TWO MEANINGS

She deposited the money by the **bank**.
She watched the water by the **bank**.

- At the input, the word has the same vector in both sentences: one table, one row per token
- After each layer the two vectors drift apart, as the words around them are folded in. By layer 11 the two Harrys are further apart than two unrelated words
- The grey line is a control: a word that means the same in both sentences also drifts, but less



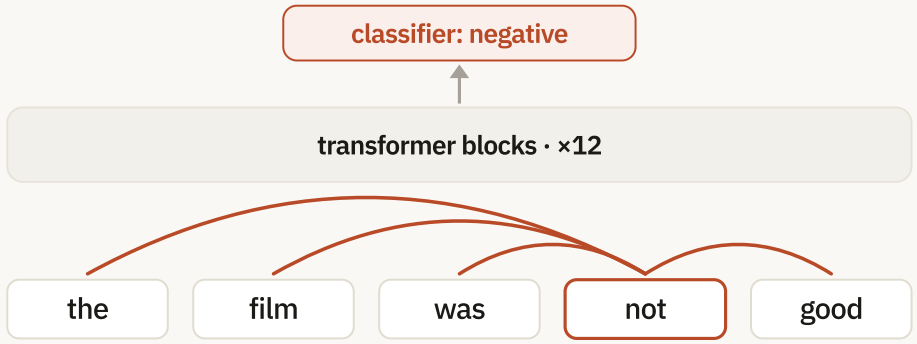
bert-base-uncased, 12 layers. Measured by the Playground's *Which Harry?* tool; the same data.

A contextual vector is a word vector that has read its sentence. That is what week 3's embeddings could not do.

PLAYGROUND Which Harry?

Two ways to read a sentence

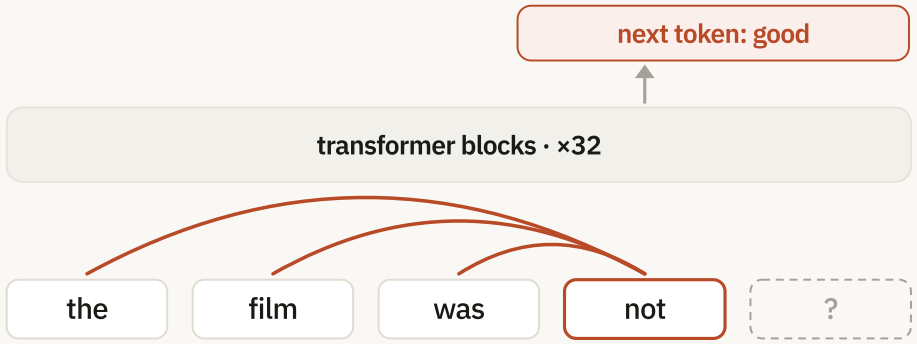
ENCODER · READS IN BOTH DIRECTIONS



BERT · Google, 2018 · RoBERTa · Meta · XLM-R · multilingual · DeBERTa · Microsoft · ModernBERT · 2024 · nomic-embed · week 3

- Every word's vector has read the whole text, including what follows it, so *not* sees *good*: what a classifier needs
- 110 to 600 million weights and one pass per text, with nothing to write. Fine-tuned on a few thousand coded texts, it often beats a prompted LLM (Bucher & Martini, 2024)

DECODER · READS LEFT TO RIGHT



GPT · OpenAI · Llama · Meta · Gemma · Google · Qwen · Alibaba · Mistral · Claude · Anthropic

- Predicting the next token, again and again, is all that writing text takes. An answer, a translation and a label are all continuations of a prompt
- Any task without training, because the task goes into the prompt. The price is billions of weights and a full pass for every token it writes

Bucher, M. J. J., & Martini, M. (2024). Fine-tuned "small" LLMs (still) significantly outperform zero-shot generative AI models in text classification. *arXiv:2406.08660*.

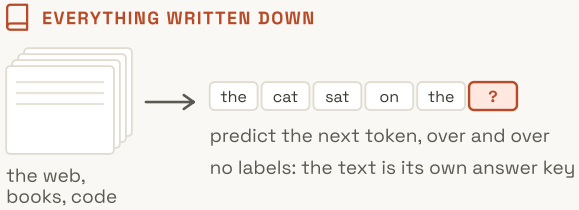
Encoder and decoder

	ENCODER · BERT (2018)	DECODER · GPT, LLAMA, CLAUDE (2018 →)
Reads	the whole text, both directions	left to right, one token at a time
Pre-trained to	fill in a masked word	predict the next token
Produces	a vector per token, or per text	the next token — and so, text
Used in research for	fine-tuned classifiers, embeddings	prompting, generation, chat
Size	110 million to a few billion weights	3 billion on a laptop to trillions in a data centre

> **Same block, two directions. The encoder is the better classifier per weight; the decoder is the one you can talk to.**

Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. *Proceedings of NAACL-HLT 2019*, 4171–4186. https://doi.org/10.18653/v1/N19-1423

From a base model to a chatbot



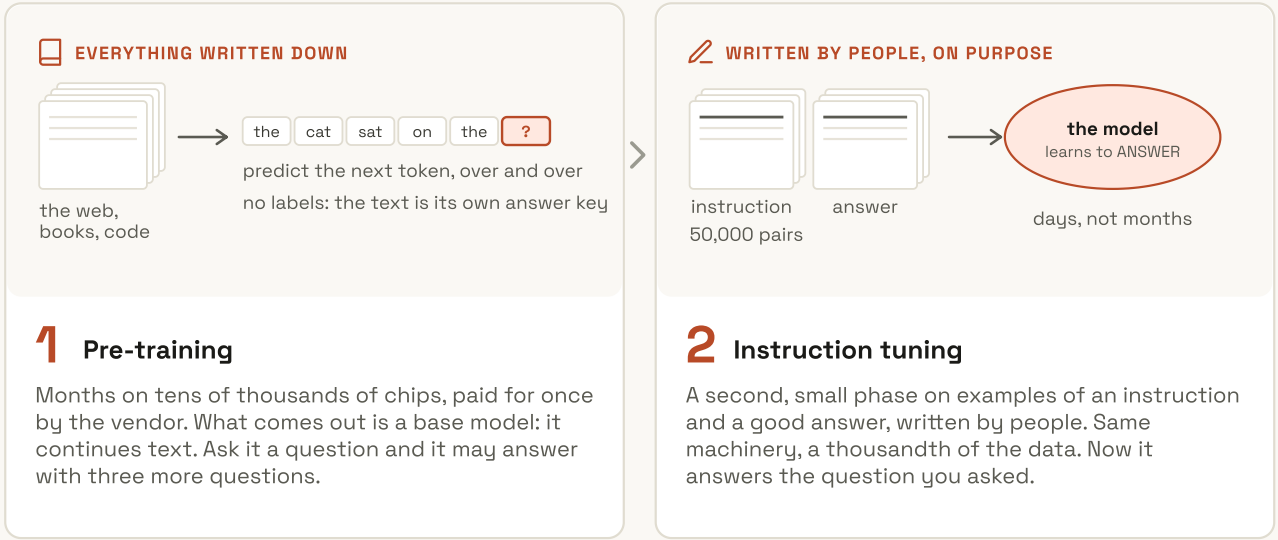
1 Pre-training

Months on tens of thousands of chips, paid for once by the vendor. What comes out is a base model: it continues text. Ask it a question and it may answer with three more questions.

1 / 5

Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., et al. (2022). Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems, 35*, 27730–27744.

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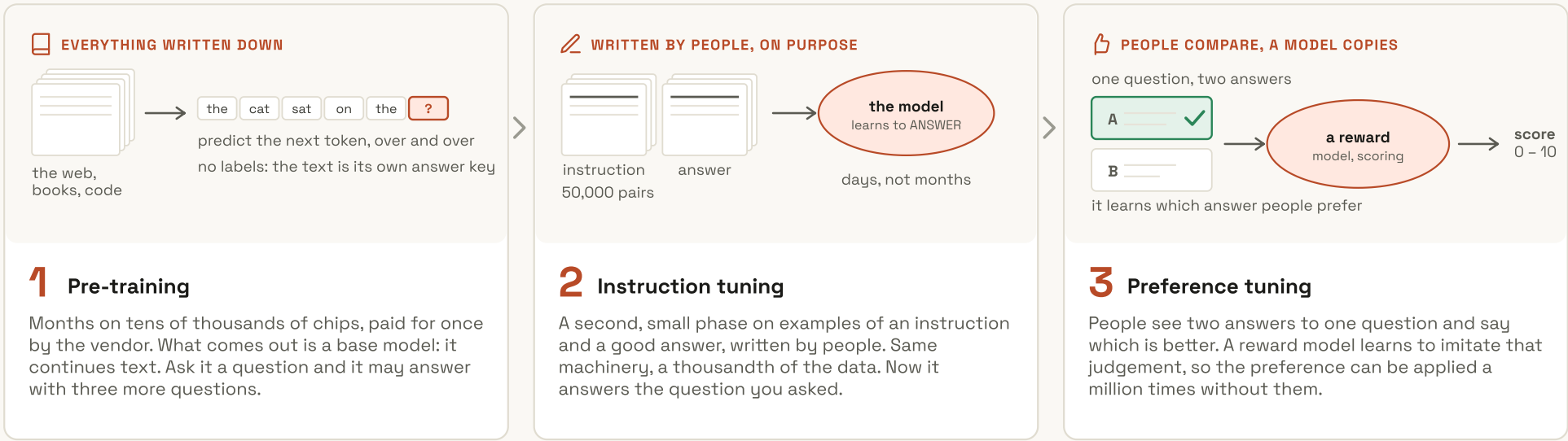
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2 Instruction tuning

A second, small phase on examples of an instruction and a good answer, written by people. Same machinery, a thousandth of the data. Now it answers the question you asked.

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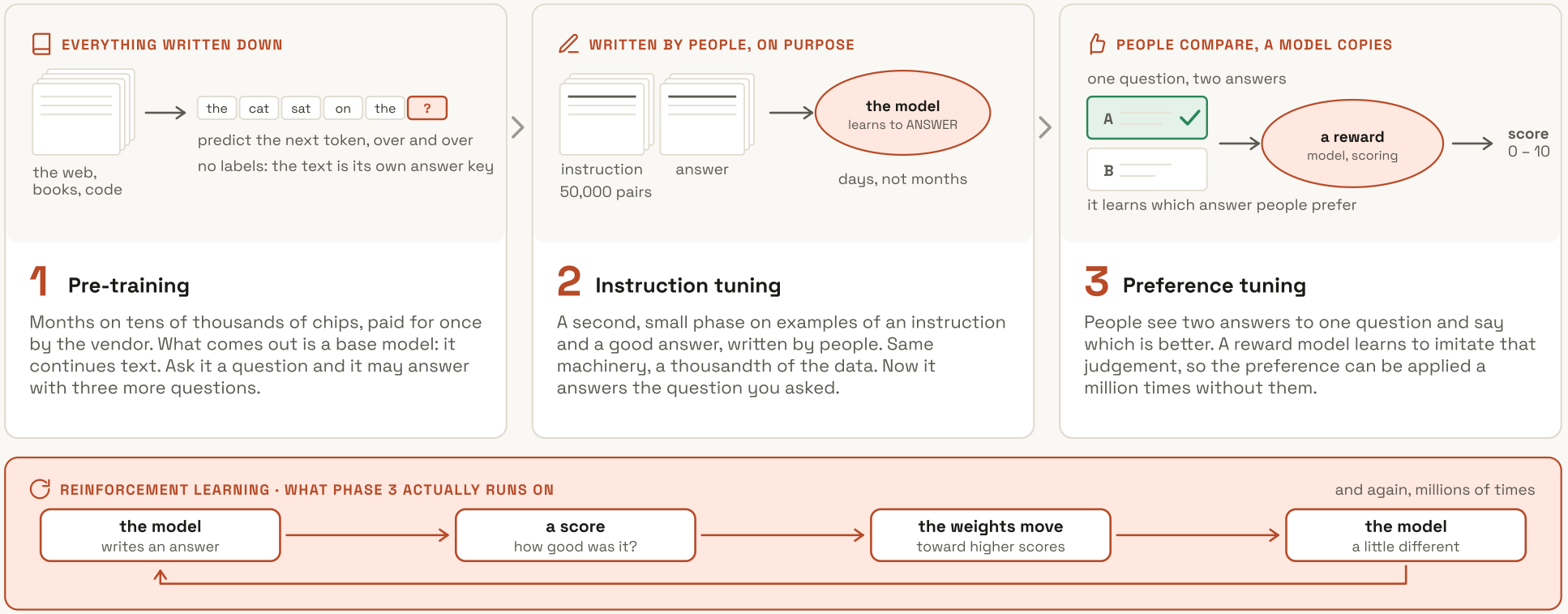
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3 Preference tuning

People see two answers to one question and say which is better. A reward model learns to imitate that judgement, so the preference can be applied a million times without them.

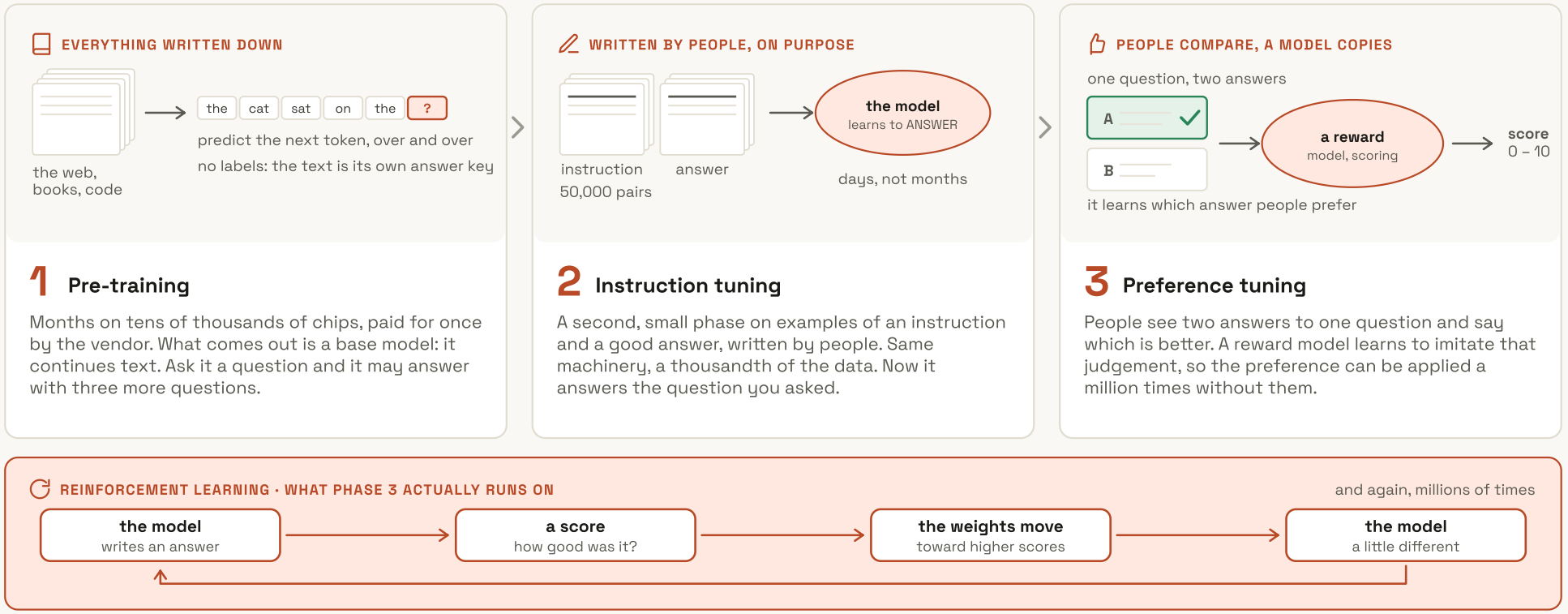
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From a base model to a chatbot



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From a base model to a chatbot



Where the score comes from is the whole difference. **From people**, it is a chat model, tuned to be liked.

From an automatic check — does the code run, is the proof right — it is a reasoning model, and nobody has to watch.

5 / 5

Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., et al. (2022). Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems, 35*, 27730–27744.

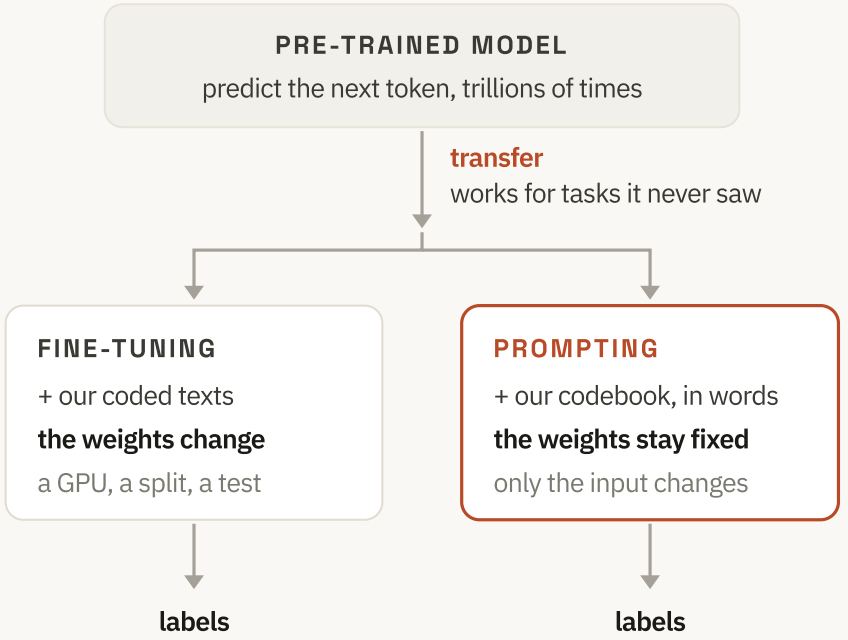
What the tuning left in the model

- The model you prompt is a **chat** model: its label on “Not good at all” is a next-token prediction shaped by all three phases
- The answer format, the hedging, the refusal to pick a side on a mixed review — trained in, by somebody else, for a general user
- Preference tuning optimises for what a rater **liked**, which is not the same as what your codebook calls correct; a fluent, agreeable answer scores well either way
- A base model, before any of this, is worse at following your instruction and *less* shaped by anyone’s idea of a good answer. Neither is neutral

> **You are prompting an instrument that was tuned to be helpful to somebody else. A coder following your codebook is something else.**

Transfer, and two ways to use it

- **Transfer learning** is the ability: what a model learned in pre-training carries over to tasks it was never trained on. Nobody trained Llama to code film reviews, and it codes them
- **Fine-tuning** uses it by training further on our coded sample, a few hundred to a few thousand texts, until the weights fit our codebook. The BERT route; it needs a GPU and week 3's split and test
- **Prompting** is simply how we talk to the model. The codebook goes into the input and the weights stay as they are. The route of this lecture
- Fine-tuning still wins on narrow tasks with enough coded data (Alizadeh et al., 2025); prompting wins on cost of entry



> **Transfer is what the pre-trained model brings. Fine-tuning and prompting are two ways to use it, and this week uses the one without a training loop.**

Alizadeh, M., Kubli, M., Samei, Z., Dehghani, S., Zahedivafa, M., Bermeo, J. D., Korobeynikova, M., & Gilardi, F. (2025). Open-source LLMs for text annotation: A practical guide for model setting and fine-tuning. *Journal of Computational Social Science, 8*(1), 17. https://doi.org/10.1007/s42001-024-00345-9

III · The models

What a large language model is

- A **transformer decoder** with billions of weights, pre-trained to predict the next token on a large share of all written text, then tuned to answer
- “Large” is doing the work: the same architecture at a hundred million weights writes plausible sentences; at a hundred billion it translates, summarises, codes and follows a codebook it has never seen
- It runs as a **service** (an API you send text to) or as **weights** you download and run yourself
- What it is not: a database, a search engine, a reasoner with a model of the world, or a coder who read your instructions carefully. It is a next-token machine that has read a great deal

> **Everything an LLM does for you is generated. Including the label, including the confidence, including the explanation.**

A language model, small enough to see



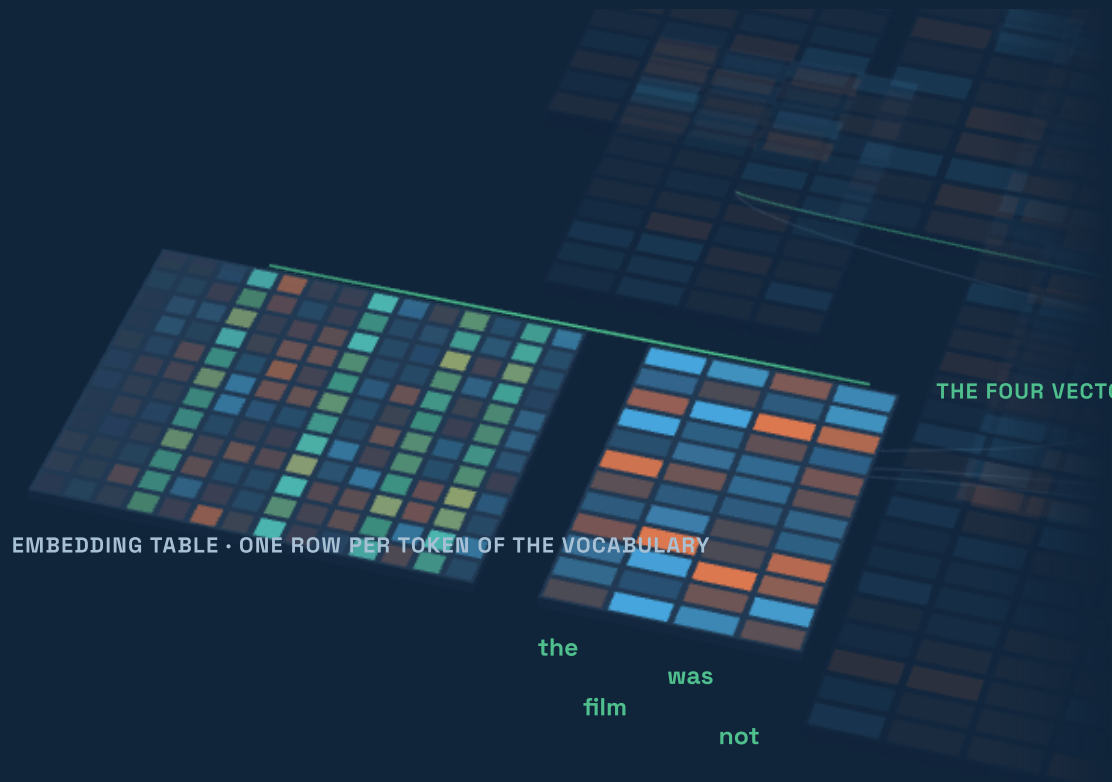
A SMALL LANGUAGE MODEL, WHOLE

Every tile is one number. Four tokens go in at the bottom, three blocks work on them, and a next token comes out at the top.

The numbers are invented; the **shapes** are a real decoder's.

After Bycroft, B. (2023). *LLM visualization*. https://bbycroft.net/llm · the numbers in the tiles are invented; the shapes are a real decoder's.

A language model, small enough to see



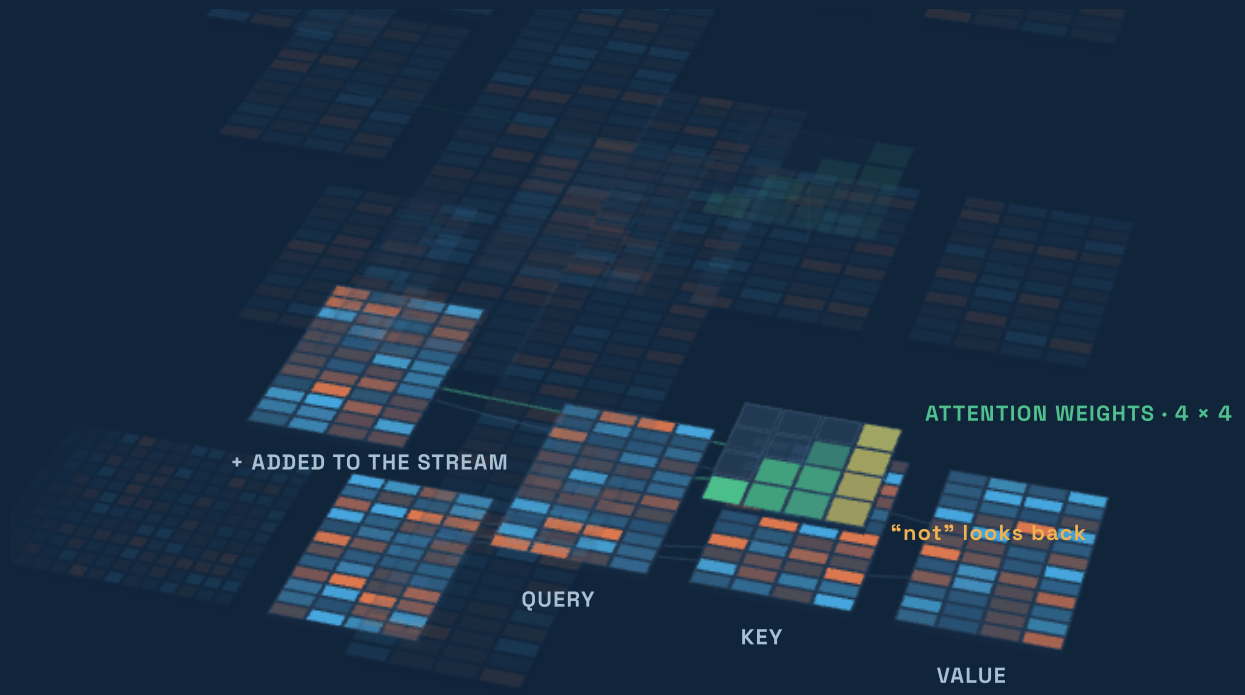
01 · TOKENS IN

Each token's row is copied out of the **embedding table**, and its position is added.

Four tokens, four vectors of twelve numbers each.

After Bycroft, B. (2023). *LLM visualization*. https://bbycroft.net/llm · the numbers in the tiles are invented; the shapes are a real decoder's.

A language model, small enough to see



02 · ATTENTION

Each vector makes a **query**, a **key** and a **value**. Queries against keys give a 4 × 4 table of weights.

Its empty half is the decoder’s rule: no token may look ahead. The weighted values are added to the stream.

After Bycroft, B. (2023). *LLM visualization*. https://bbycroft.net/llm · the numbers in the tiles are invented; the shapes are a real decoder’s.

A language model, small enough to see



03 · FEED-FORWARD

Then each vector goes through a network **four times as wide**, on its own, and what comes out is added to it.

After Bycroft, B. (2023). *LLM visualization*. https://bbycroft.net/llm · the numbers in the tiles are invented; the shapes are a real decoder's.

A language model, small enough to see

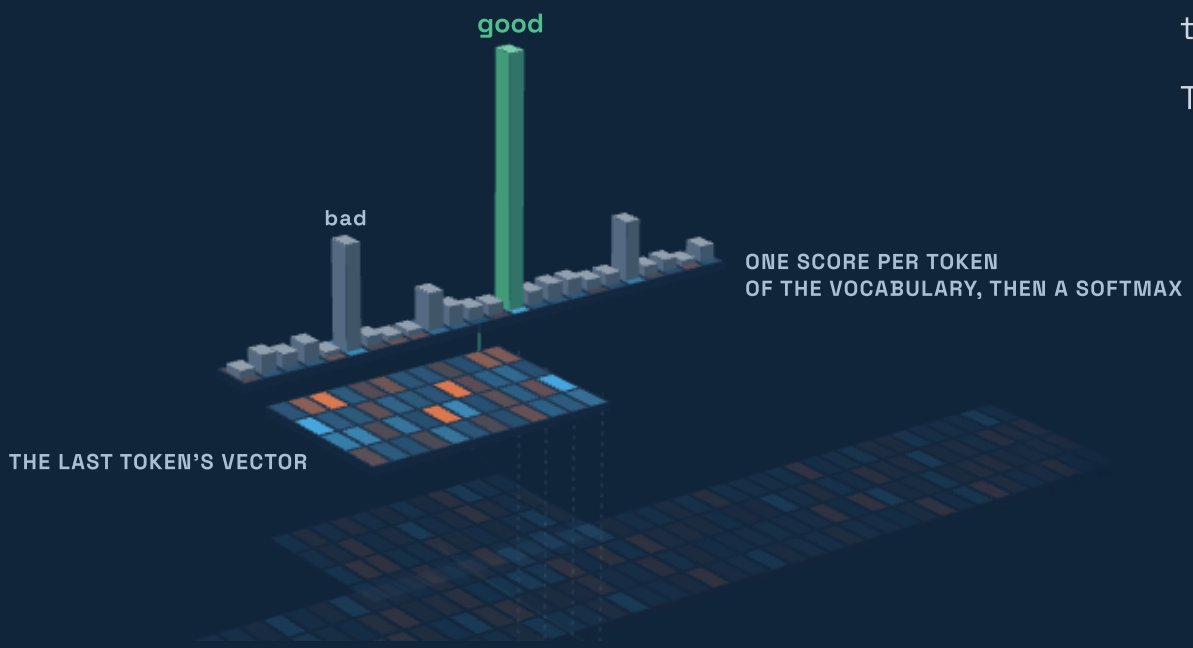


04 · THE STACK

Attention and feed-forward make one **block**. The stream of vectors rises through all three, and every block adds to it.

After Bycroft, B. (2023). *LLM visualization*. https://bbycroft.net/llm · the numbers in the tiles are invented; the shapes are a real decoder's.

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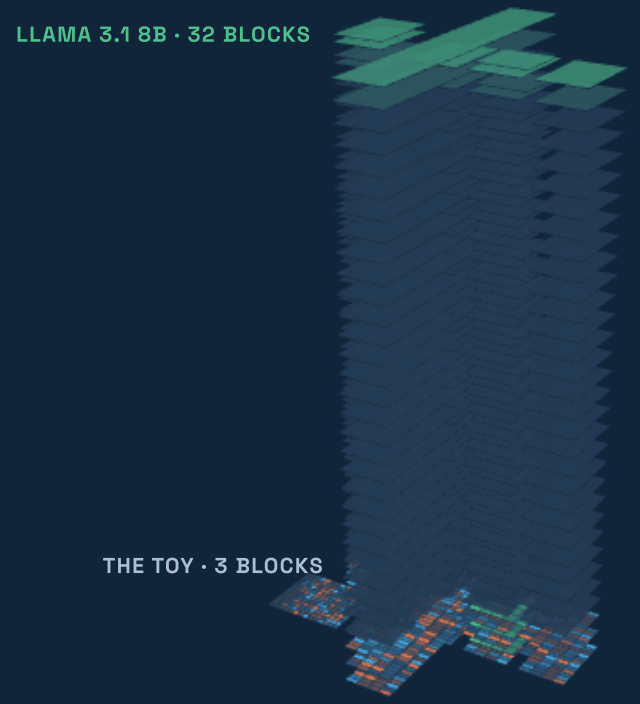
05 · THE NEXT TOKEN

At the top only the **last** vector is read: one score for every token in the vocabulary, then a **softmax**.

The tallest bar is the prediction: **good**.

After Bycroft, B. (2023). *LLM visualization*. https://bbycroft.net/llm · the numbers in the tiles are invented; the shapes are a real decoder's.

A language model, small enough to see



06 · THE REAL SIZE

Llama 3.1 8B is the same machine with **32 blocks**, 4,096 numbers per token instead of 12, and 128,256 tokens in its vocabulary.

About 8 billion weights. This toy has about 6,000.

< 7 / 7 >

After Bycroft, B. (2023). *LLM visualization*. https://bbycroft.net/llm · the numbers in the tiles are invented; the shapes are a real decoder's.

The models, September 2026

Closed

weights secret · used through the vendor's API

OPENAI
GPT-6 Astra
Sep 2026 · GPT-5.6 Sol before it

ANTHROPIC
Claude Fable 5.1
Sep 2026 · Opus 5, Sonnet 5, Haiku 4.5

GOOGLE
Gemini 3.8
Sep 2026 · Flash and Pro tiers

xAI
Grok 4.6
Aug 2026

Open weights

download the weights · run them anywhere, or rent them

On this laptop

open weights, shrunk, run through Ollama · this course

SEPTEMBER 2026 · THE NAMES CHANGE MONTHLY; THE COLUMNS DO NOT

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Llama 4
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DeepSeek V4
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Aug 2026 · Qwen 2.5 runs in this deck

MISTRAL · GOOGLE · MOONSHOT
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META
`llama3.2:3b`
2.0 GB · what a student laptop runs

META
`llama3.1:8b`
4.9 GB · the cold open's model

GOOGLE
`gemma3:12b` · `gemma4:26b`
8.1 GB · 17 GB

ALIBABA
`qwen2.5:32b`
19 GB · the largest this deck ran

SEPTEMBER 2026 · THE NAMES CHANGE MONTHLY; THE COLUMNS DO NOT

The models, September 2026

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Sep 2026 · Flash and Pro tiers

xAI
Grok 4.6
Aug 2026

WHERE THE TEXT GOES
to the vendor's servers

CAN THE VERSION BE PINNED?
only until it is retired

WHAT A TEXT COSTS
cents, per token

Open weights

download the weights · run them anywhere, or rent them

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MISTRAL · GOOGLE · MOONSHOT
Mistral Medium 3.5 · Gemma 4 · Kimi K2
2025 to 2026 · some open, some not

WHERE THE TEXT GOES
to whoever hosts it

CAN THE VERSION BE PINNED?
yes, forever

WHAT A TEXT COSTS
cents, or GPU hours

On this laptop

open weights, shrunk, run through Ollama · this course

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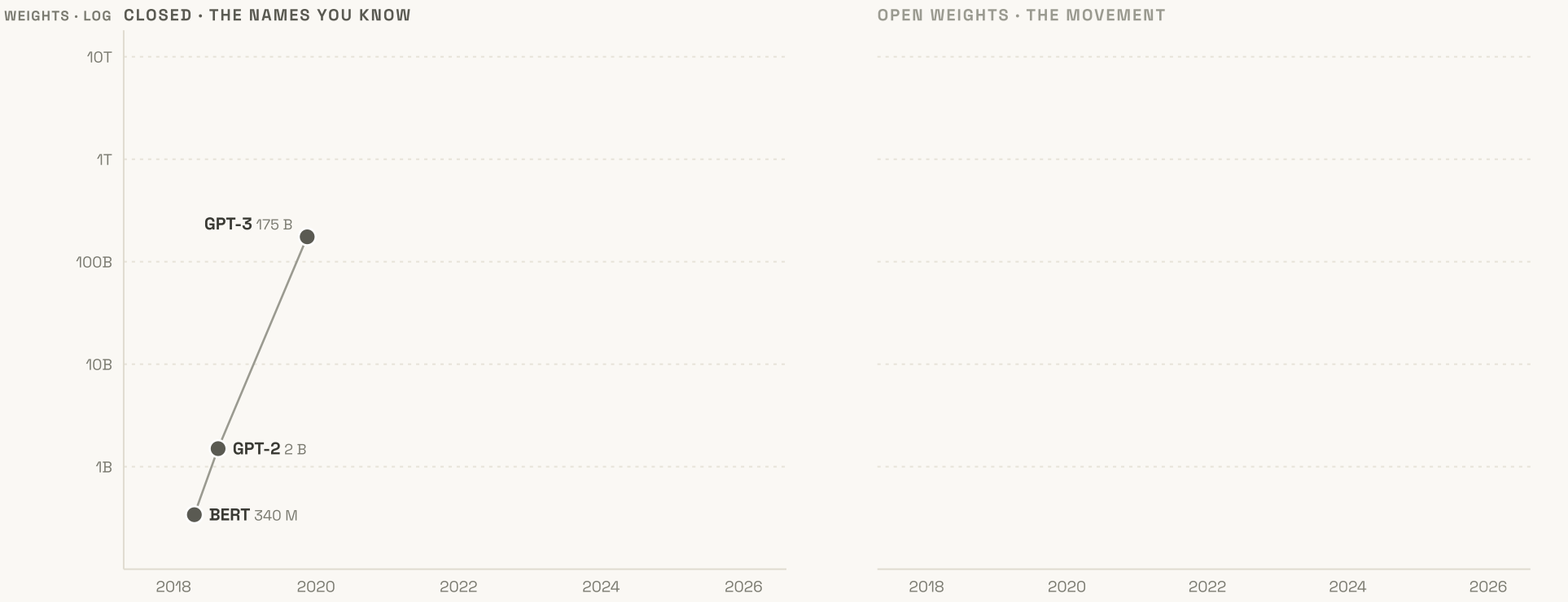
WHERE THE TEXT GOES
nowhere: it stays on the laptop

CAN THE VERSION BE PINNED?
yes, forever

WHAT A TEXT COSTS
electricity, and time

SEPTEMBER 2026 · THE NAMES CHANGE MONTHLY; THE COLUMNS DO NOT

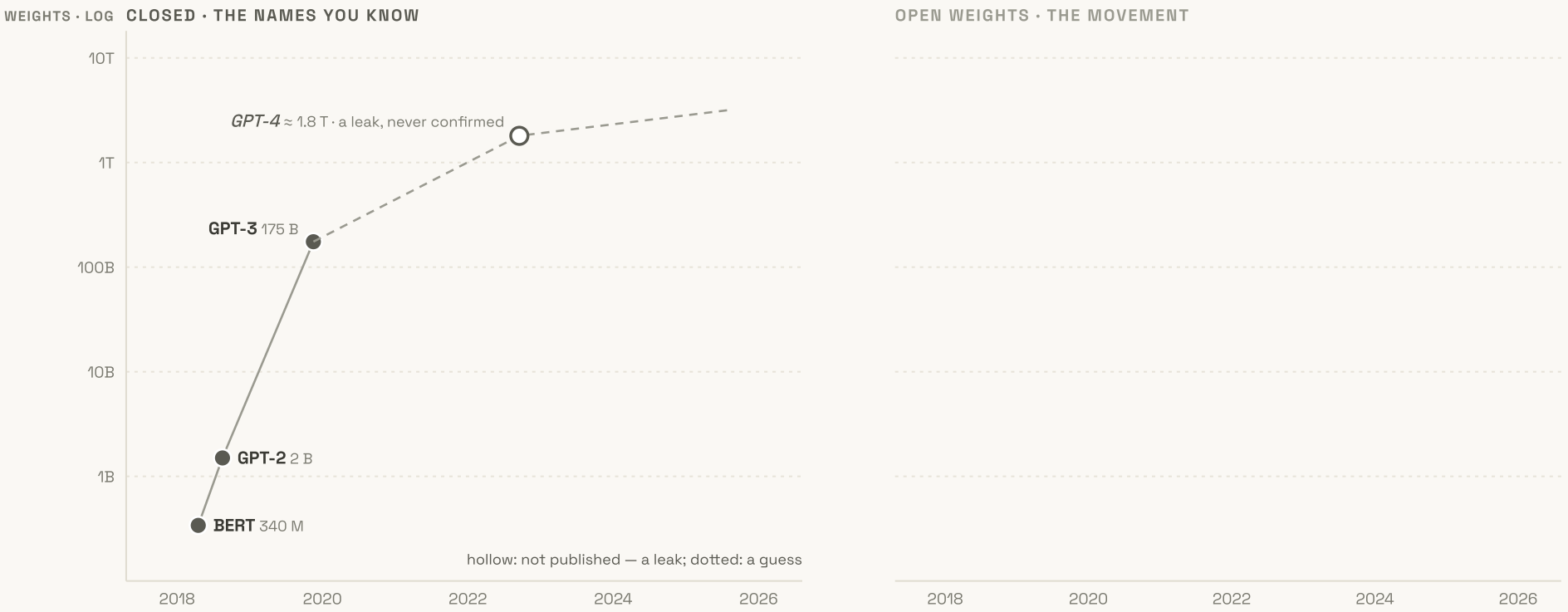
Bigger every year



1 / 6

Kaplan, J., et al. (2020). Scaling laws for neural language models. *arXiv:2001.08361* · Ben Abacha, A., et al. (2024). MEDEC: A benchmark for medical error detection and correction in clinical notes. *arXiv:2412.19260* (the 2024 parameter estimates) · GPT-4 and Grok figures: leak (SemiAnalysis, 2023) and xAI statements (2025), unconfirmed.

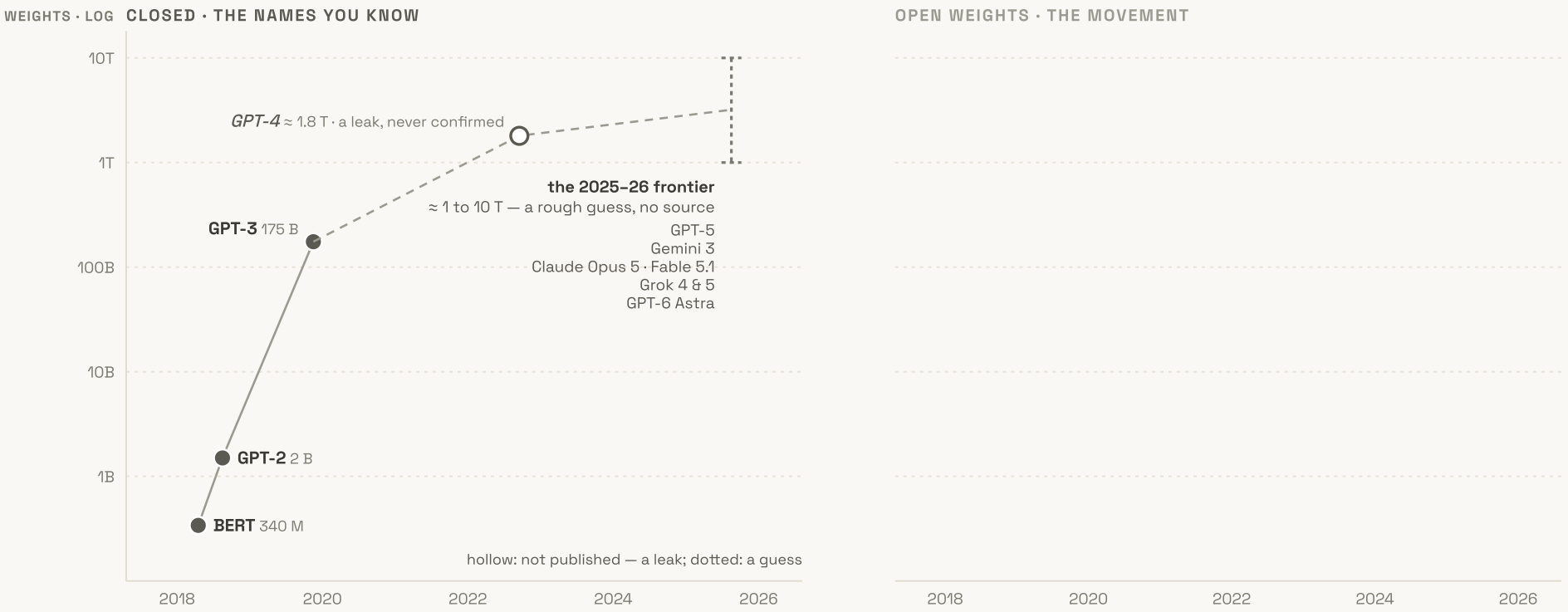
Bigger every year



2 / 6

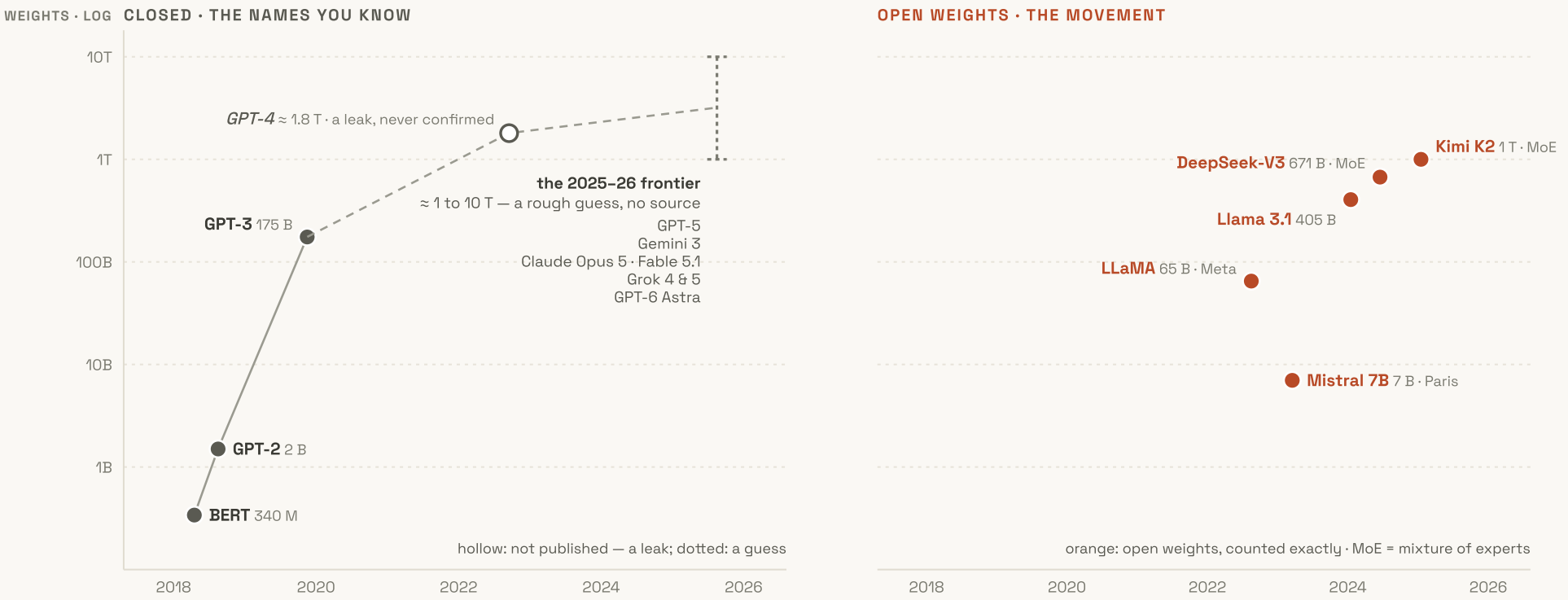
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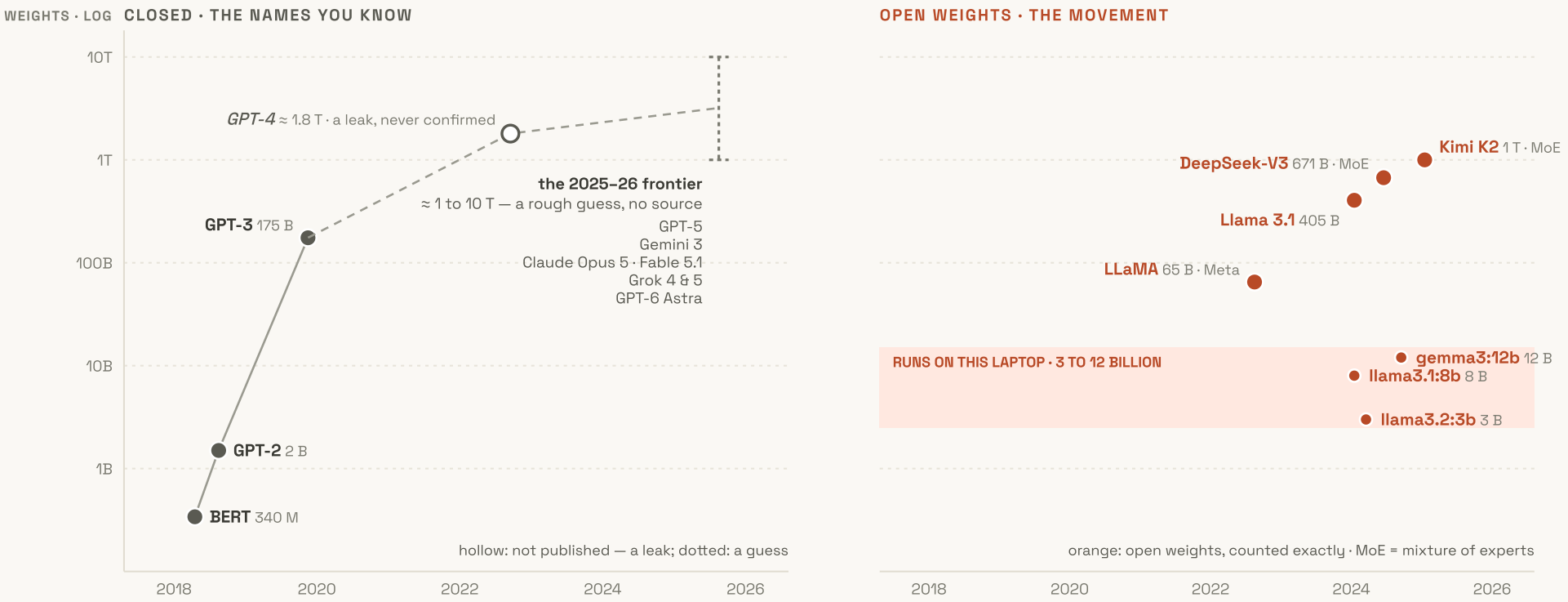
Bigger every year



4 / 6

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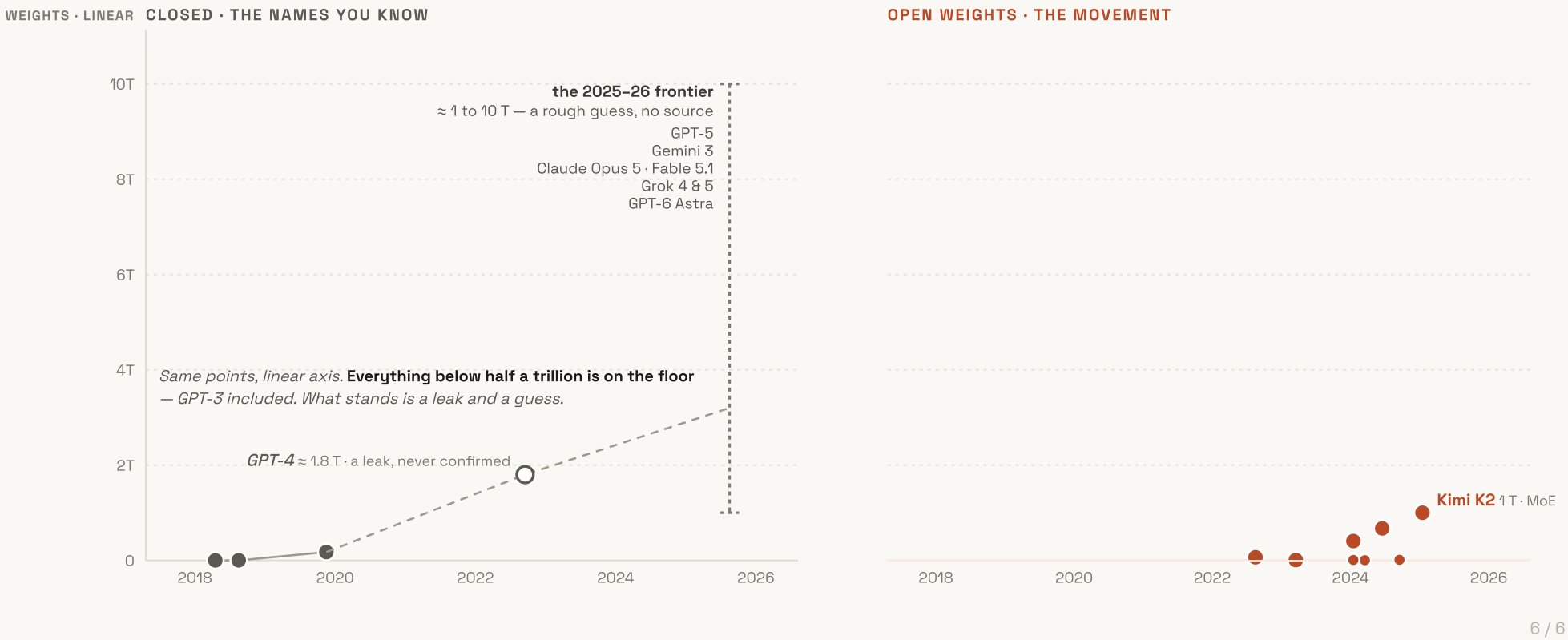
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Talking to a model from R

```r
library(tidyllm)

llm_message("In two sentences: what is a codebook
             in content analysis?") |>
  chat(ollama(.model = "llama3.1:8b",
              .temperature = 0)) |>
  get_reply()
```

- `llm_message()` builds the conversation; `chat()` sends it to a provider — here a local Ollama server, but `openai()`, `claude()` or `gemini()` slot in unchanged
- `.temperature = 0`: take the most probable token every time
- `get_reply()` pulls the answer out as a string

OUTPUT

```
In content analysis, a codebook is a detailed
guide that outlines the specific codes,
categories, and definitions used to analyze and
interpret the content of texts, images, or other
data. The codebook serves as a reference point for
researchers to ensure consistency and reliability
in their coding process, and it typically includes
information on code definitions, examples, and
rules for applying the codes to the data.
```

> A model that can define a codebook can be handed one. That is the whole trick of Part IV.

Break

> Ten minutes. After the break: the prompt as an instrument, and what moves its score.

IV · LLMs as instruments

Deductive, inductive — and instructed

	DICTIONARY · WEEK 2	CLASSIFIER · WEEK 3	LANGUAGE MODEL · WEEK 4
The rules come from	you, as a word list	coded examples, learned	you, as a codebook in words
What the machine reads	listed words, one by one	the features of coded texts	the whole text, in context
Coded texts needed	none	thousands, split in two	none — or a handful, as examples
Scored against	the gold standard	the held-out part of it	the gold standard, all of it
Logic	deductive	inductive	deductive rules, inductive machine

> **The codebook is back in your hands, as in week 2. The reader is the inductive machine of week 3. Validation is unchanged since week 2.**

The steps, with a language model

n = 200



coded by people, with a codebook — as before

1 Hand-code a sample

Unchanged. A set of texts coded by people with a codebook is still the only place meaning enters — and the only thing the model is scored against.

The steps, with a language model



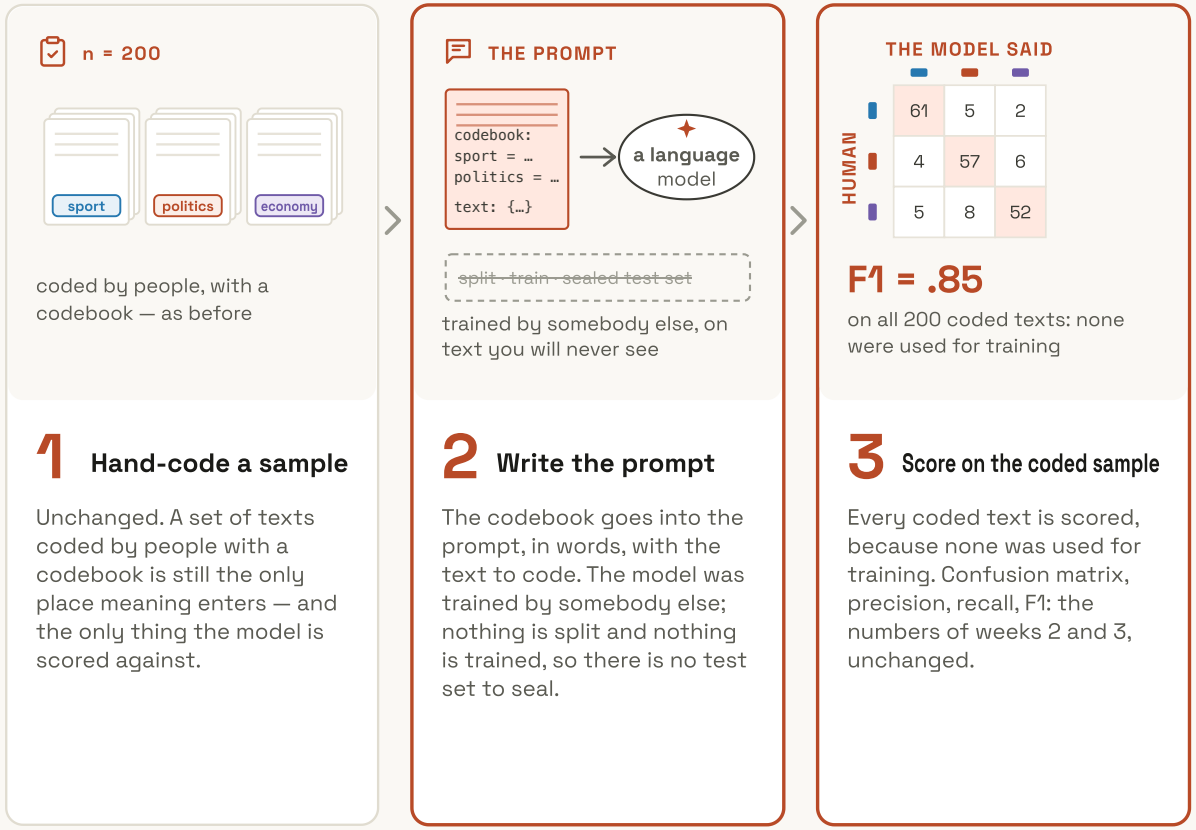
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2 Write the prompt

The codebook goes into the prompt, in words, with the text to code. The model was trained by somebody else; nothing is split and nothing is trained, so there is no test set to seal.

The steps, with a language model



1 Hand-code a sample

Unchanged. A set of texts coded by people with a codebook is still the only place meaning enters — and the only thing the model is scored against.

2 Write the prompt

The codebook goes into the prompt, in words, with the text to code. The model was trained by somebody else; nothing is split and nothing is trained, so there is no test set to seal.

3 Score on the coded sample

Every coded text is scored, because none was used for training. Confusion matrix, precision, recall, F1: the numbers of weeks 2 and 3, unchanged.

The steps, with a language model

n = 200

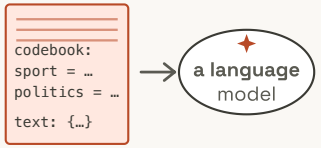


coded by people, with a codebook — as before

1 Hand-code a sample

Unchanged. A set of texts coded by people with a codebook is still the only place meaning enters — and the only thing the model is scored against.

THE PROMPT



~~split · train · sealed test set~~

trained by somebody else, on text you will never see

2 Write the prompt

The codebook goes into the prompt, in words, with the text to code. The model was trained by somebody else; nothing is split and nothing is trained, so there is no test set to seal.

THE MODEL SAID

HUMAN			
	61	5	2
	4	57	6
	5	8	52

F1 = .85

on all 200 coded texts: none were used for training

3 Score on the coded sample

Every coded text is scored, because none was used for training. Confusion matrix, precision, recall, F1: the numbers of weeks 2 and 3, unchanged.

n = 100,000



labelled by the model: 100,000 prompts

4 Apply to the corpus

Only if the score is acceptable. Every text is one prompt, so the cost is per text, in seconds or in cents — and the model must be the same one that was scored.

The steps, with a language model

n = 200

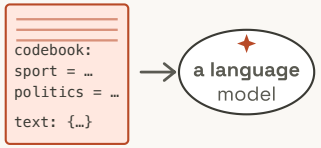


coded by people, with a codebook — as before

1 Hand-code a sample

Unchanged. A set of texts coded by people with a codebook is still the only place meaning enters — and the only thing the model is scored against.

THE PROMPT



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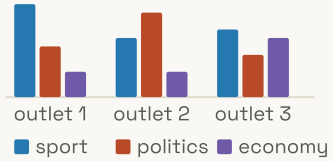
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labelled by the model: 100,000 prompts

4 Apply to the corpus

Only if the score is acceptable. Every text is one prompt, so the cost is per text, in seconds or in cents — and the model must be the same one that was scored.



with F1 = .85, the prompt and the model beside it

5 Analyse the labels

The predicted codes become a variable, as before. The validation figures travel with them — and now the prompt and the model version do too.

5 / 5

One classification

```
codebook <- "You are coding sentences from restaurant reviews for a
  content analysis. Codebook: positive = the sentence expresses a
  favourable evaluation of the restaurant, its food, staff or
  experience; negative = an unfavourable evaluation.
  Code the following sentence.\n\nSentence: {text}"

sentence <- yelp_test$text[1]

llm_message(str_replace(codebook, stringr::fixed("{text}"), sentence)) |>
  chat(ollama(.model = "llama3.1:8b", .temperature = 0)) |>
  get_reply()
```

- The prompt has three parts: the role, the **codebook** with a definition per category, and the text
- The same wording every time; only the text changes

OUTPUT

```
sentence: Stopped by during the late May bank holiday
off Rick Steve recommendation and loved it.
```

```
Based on the codebook, I would code this
sentence as: **Positive** Reasoning: The
sentence expresses a favourable evaluation
of the restaurant, as indicated by the word
"loved it". The fact that the reviewer stopped
by based on a recommendation from Rick Steve
also suggests a positive predisposition
towards the restaurant.
```

```
human label: positive
```

> A label, and usually a sentence around it. A reply is text; a variable is not. That gap is the next slide.

Structured output

```
schema <- tidyllm_schema(
  name       = "coding",
  label      = field_fct("the code", .levels = c("positive", "negative")),
  confidence = field_dbl("how sure you are, 0 to 1"),
  reason     = field_chr("one short sentence"))

llm_message(str_replace(codebook, stringr::fixed("{text}"), sentence)) |>
  chat(ollama(.model = "llama3.1:8b", .temperature = 0,
              .json_schema = schema)) |>
  get_reply_data()
```

- A **schema** declares the fields and the allowed values; the server constrains the decoding so the reply *cannot* be anything else
- `get_reply_data()` returns an R list: the label is a factor level, not a sentence to parse
- The confidence and the reason are generated text too. Useful to read; not measurements
- Note the **order of the fields**. The model fills them in the order you declare, so a `reason` before the `label` lets it write before it commits. In this deck's own runs one model needed exactly that

OUTPUT

```
$label      positive
$confidence 0.9
$reason     The reviewer mentions that they
'loved it', which is a strong positive
sentiment.
```

> Structured output ends parse failures: twelve in this deck's free-text runs, none under a schema. It does not end wrong answers: in the paper in Part V, the reply format alone moves small models by up to twelve points.

Two hundred sentences

```r
schema <- tidyllm_schema(name = "coding",
  label = field_fct("the code", .levels = c("positive", "negative")))

code_one <- function(text) {
  llm_message(str_replace(codebook, stringr::fixed("{text}"), text)) |>
    chat(ollama(.model = "llama3.1:8b", .temperature = 0, .seed = 42,
                .json_schema = schema)) |>
    get_reply_data() |> pluck("label")
}

coded <- yelp_test |>
  mutate(pred = map_chr(text, code_one, .progress = TRUE),   # ~100 s
         pred = factor(pred, levels = levels(label)))
```

- One function, mapped over the texts: the loop is the study
- Two hundred sentences take a hundred seconds on this laptop with the 8-billion-weight model; the cached result is what the next slides score

OUTPUT

```
# A tibble: 6 × 3
  text                              label    pred
  <chr>                             <fct>    <fct>
1 Stopped by during the late ...    positive positive
2 A great touch.                    positive positive
3 Service was very prompt.          positive positive
4 The cashier had no care wha...    negative negative
5 I was disgusted because I w...    negative negative
6 The Burrittos Blah!               negative negative
```

> **Every row is a prompt. Every prompt is a full pass through the model. There is no fitted object to keep.**

Validate: the numbers of weeks 2 and 3

DATA · 200 HELD-OUT YELP SENTENCES

```
coded |> conf_mat(truth = label, estimate = pred)

coded |> class_metrics(truth = label, estimate = pred)
```

- The same `conf_mat()` and the same metric set as last week: nothing about validation is new
- All two hundred coded sentences are test cases, because none were used to train anything

> A number this high on Yelp sentences is not a surprise; a number this high on *your* construct would be. Score it before you believe it.

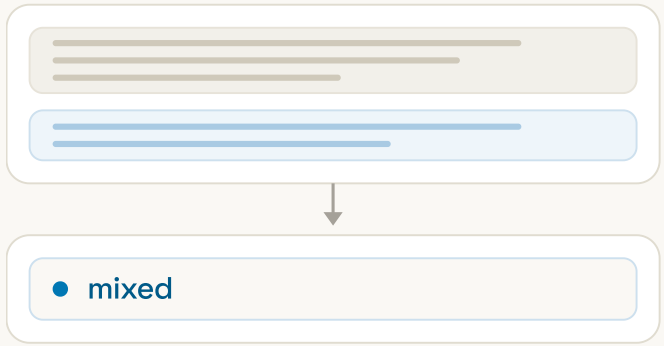
OUTPUT

```
          Truth
Prediction positive negative
  positive       99        2
  negative        1       98
```

```
# A tibble: 4 × 2
  .metric   .estimate
  <chr>         <dbl>
1 accuracy      0.985
2 precision     0.98
3 recall        0.99
4 f_meas        0.985
```

Zero-shot, few-shot, chain of thought

Zero-shot
the codebook and the text



Few-shot
a handful of coded examples added



Chain of thought
reasoning asked for, then the label



codebook the text to code reasoning, asked for and given back

> Only the middle of the prompt changes: few-shot adds to what goes in, chain of thought to what comes back. Which of them helps is measured on the next slides, on forty-eight texts.

The prompt as codebook

```
Label the review as positive, negative, or mixed.
Use 'mixed' whenever the review contains BOTH substantial praise AND substantial
criticism, even if one outweighs the other.
```

```
# few-shot: the examples appended for this definition
```

```
The effects are dated, but the story still lands. → mixed
Superb cast, wasted on a script that goes nowhere. → mixed
Gorgeous from start to finish. → positive
```

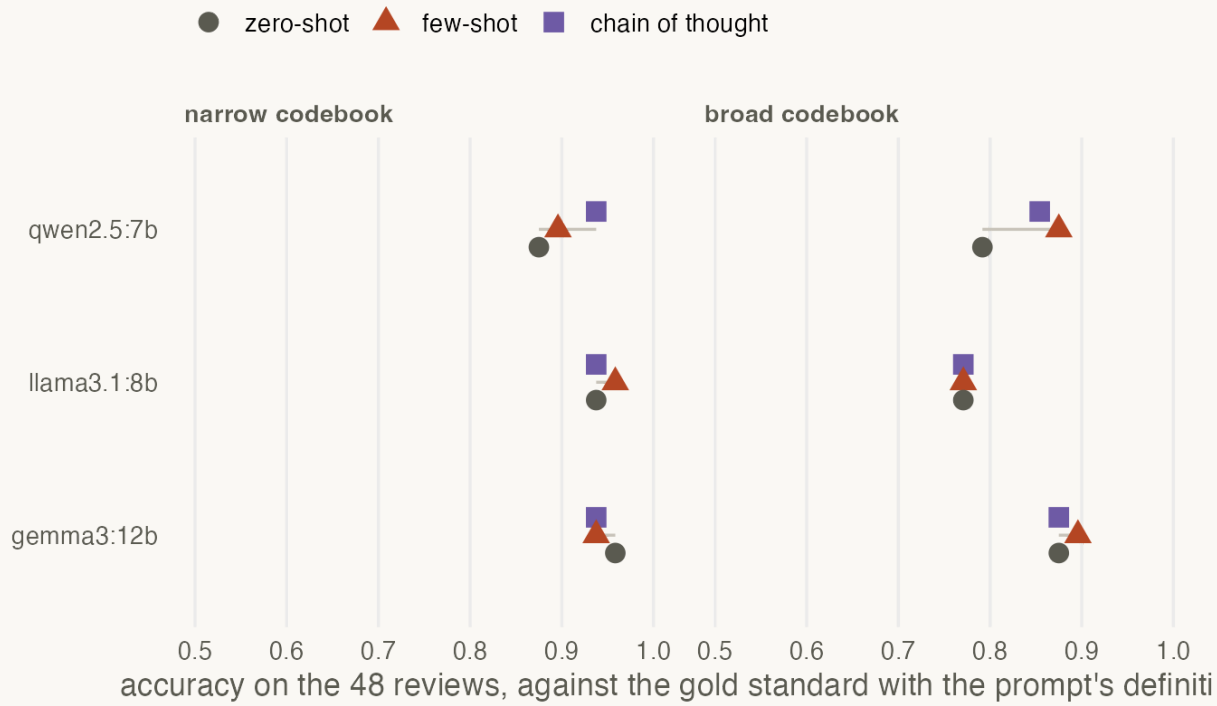
- The Playground's prompt lab: forty-eight film reviews, hand-coded on three fields, from which two codebooks are derived — a **broad** *mixed* (any praise and criticism together) and a **narrow** one (only when the verdict is genuinely unclear)
- They disagree on **13 of the 48** texts: the borderline ones
- Every definition was run zero-shot, few-shot and with chain of thought, on three local models, at temperature 0

> **A prompt is a codebook. Two defensible codebooks give two different gold standards — before any model is involved.**

PLAYGROUND The prompt is the instrument

Does few-shot help? Does reasoning?

DATA · 48 FILM REVIEWS, HAND-CODED FOR THIS COURSE



- Three models, two codebooks, three techniques: eighteen scores on the same forty-eight texts
- No technique wins everywhere. Few-shot helps one model on one codebook and hurts another; chain of thought costs ten times the time for gains you would not bet on
- The differences between *models* are as large as the differences between *techniques*

> **“Few-shot helps” is a hypothesis about your task. Test it on your gold set; do not import it.**

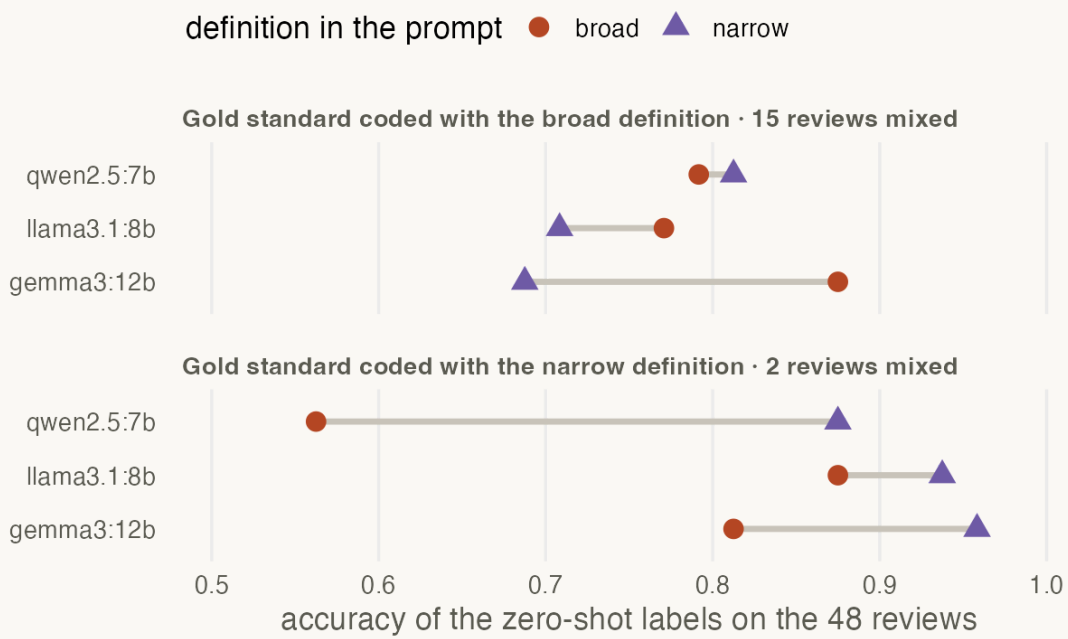
PLAYGROUND · The prompt is the instrument

The codebook moves the score

DATA · THE SAME 48 FILM REVIEWS, TWO DERIVED CODEBOOKS

REVIEW	BROAD GOLD	NARROW GOLD
Slow, but worth it.	mixed	positive
Well acted, but pointless.	mixed	negative
Gorgeous cinematography throughout.	positive	positive

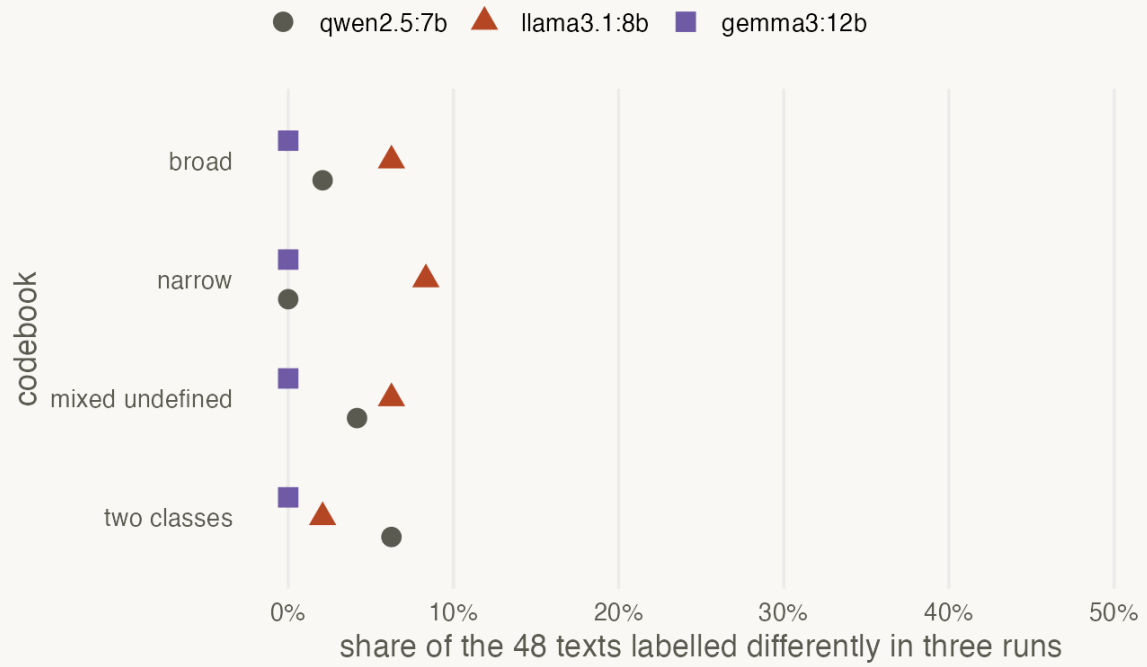
- One hand-coding, two gold standards. **Broad** calls a review *mixed* whenever it has praise and criticism; **narrow** only when it reaches no verdict. They differ on 13 of the 48 reviews, like the first two rows
- Each prompt gave the model one of the two definitions. A panel holds one gold standard fixed and compares the two prompts on it



> Against either gold standard, the prompt with the same definition scored higher in 5 of 6 comparisons, by up to 31 points. The prompt has to carry the definition the coders used.

Run to run

DATA · THE SAME 48 REVIEWS, THREE PASSES AT TEMPERATURE 0.8



- The same prompt, three times, at temperature 0.8: the model samples from its distribution instead of taking the top token
- Up to a fifth of the texts change label between runs — the borderline ones, again
- At temperature 0 with a fixed seed a *local* model is deterministic. An API model is not guaranteed to be, and a retired version is gone

> **A coder who changes a fifth of the labels on a second pass is not reliable. Neither is a prompt. Measure it the same way.**

Reiss, M. V. (2023). Testing the reliability of ChatGPT for text annotation and classification: A cautionary remark. *arXiv:2304.11085*.

What to report

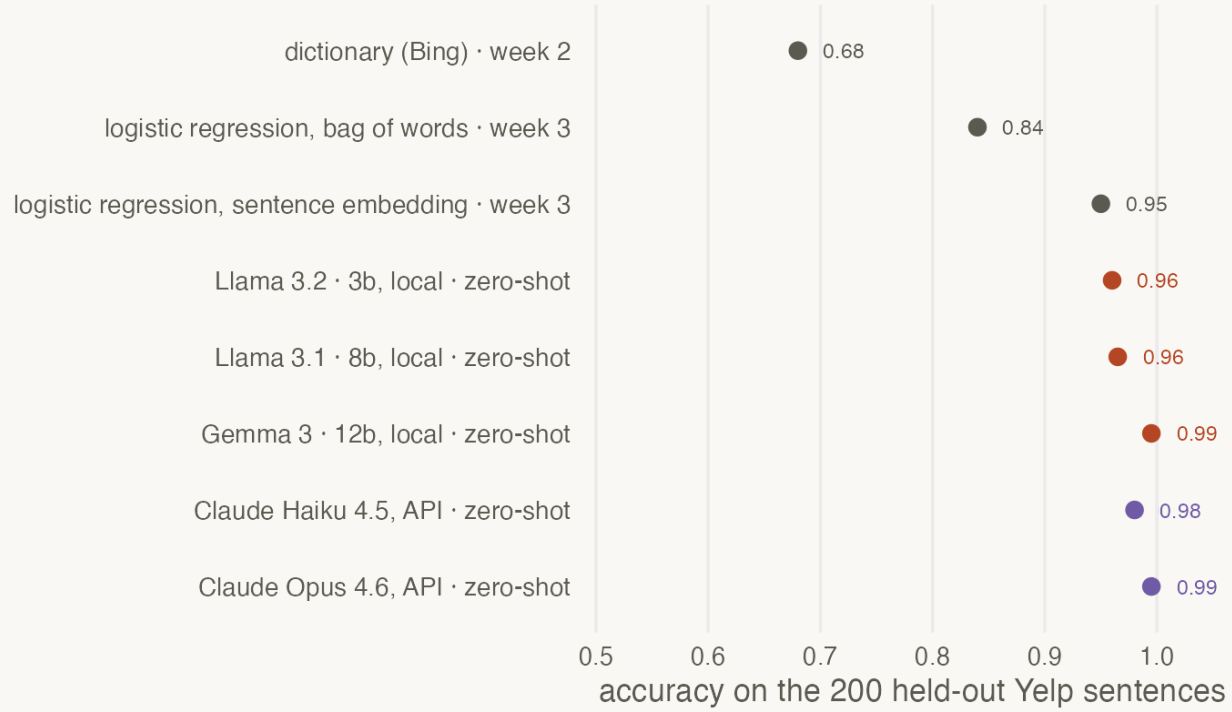
- **The prompt, verbatim**, including the codebook and any examples — it is the instrument
- **The model and its version**, the provider, the date of the run, temperature and seed
- **A validation on a hand-coded gold set**, with the metric that fits the class balance — and the humans' own agreement on the same texts
- **Sensitivity:** at least two prompt wordings and at least two models, with the scores side by side; and run-to-run agreement if the temperature was above zero
- **Parse failures**, and what was done with them
- **Where the text went**, and the raw replies archived

> Six lines that turn a prompt into a method section. Lab A asks for every one of them.

V · Four generations, one gold set

Four generations, one gold set

DATA · 200 HELD-OUT YELP SENTENCES · MODELS IN FREE TEXT



Every row is scored on the same 200 held-out sentences of Kotzias et al.'s Yelp set. The dictionary's silences count as errors; the two classifiers were trained on the other 800 sentences; the language models saw none of them.

- The 2013 dictionary, the 2019 classifier, the 2023 embedding, the 2024–25 language models: each generation beats the last on the same sentences
- Every local model here, down to the three-billion one on a student laptop, beats last week's best classifier — with no training data at all
- The two bottom rows are **frontier models through an API**: Haiku .98, Opus .995 — and .995 is where the twelve-billion model on your laptop already sits. Sentence 145, two slides on, is why nobody scores higher

> **This is the figure the course was built toward. The rest of Part V is why it is not the end.**

The numbers you report

DATA · 200 HELD-OUT YELP SENTENCES · MODELS IN FREE TEXT

METHOD	ACCURACY	PRECISION +	RECALL +	F1 +	PRECISION −	RECALL −	F1 −
dictionary (Bing) · week 2	**0.680**	**0.627**	0.890	0.736	0.810	**0.470**	**0.595**
logistic regression, bag of words · week 3	0.840	0.847	0.830	0.838	0.833	0.850	0.842
logistic regression, sentence embedding · week 3	0.950	0.969	0.930	0.949	0.933	0.970	0.951
Llama 3.2 · 3b, local · zero-shot	0.960	1.000	0.920	0.958	0.926	1.000	0.962
Llama 3.1 · 8b, local · zero-shot	0.965	0.989	0.940	0.964	0.943	0.990	0.966
Gemma 3 · 12b, local · zero-shot	0.995	1.000	0.990	0.995	0.990	1.000	0.995
Claude Haiku 4.5, API · zero-shot	0.980	0.980	0.980	0.980	0.980	0.980	0.980
Claude Opus 4.6, API · zero-shot	0.995	1.000	0.990	0.995	0.990	1.000	0.995

Precision, recall and F1 for each class: + the positive class, − the negative; values below .70 in colour. The dictionary's .89 recall on positives is bought with .47 on negatives — it calls most things positive. The language-model rows are the free-text reply, as in the figure.

The one sentence every model got wrong

DATA · 200 HELD-OUT YELP SENTENCES · SENTENCE 145

> “Nice blanket of moz over top but i feel like this was done to cover up the subpar food.”

THE GOLD STANDARD SAYS

positive

One label, assigned once when Kotzias et al. built the sentence set in 2015; used by hundreds of papers since, and by this course since week 2. Nobody checked it — until fifteen coders disagreed with it at once.

ALL 15 CELLS SAY

negative

Five models, three reply formats, every single one. Claude Opus, asked to give its reason before its label:

“While the sentence acknowledges a positive element (nice blanket of mozzarella), the overall sentiment is negative, suggesting the cheese was used to disguise subpar food quality.”

Kotzias, D., Denil, M., de Freitas, N., & Smyth, P. (2015). From group to individual labels using deep features. *KDD ’15*, 597–606 — the “Sentiment Labelled Sentences” set (UCI). Sentence 145 of the course’s 200 held-out sentences; all fifteen model cells from results/llm2.

Last week's table, four rows longer

DATA · 300 HELD-OUT ARXIV ABSTRACTS · MODELS IN FREE TEXT



- The arXiv abstracts of week 3, three disciplines, a hundred held-out abstracts each — a harder task: long texts, and *statistics* overlaps with both neighbours
- The three local models land within two points of each other, **.74 to .76**, and the frontier models at .76 and .78 — all of them still lean toward *computer science* for anything with an algorithm in it
- The classifier trained on 4,200 coded abstracts **beats every one of them** at .83 — with coded data and a narrow construct, training still wins

> **Zero-shot is not free accuracy. With coded data and a narrow construct, last week's classifier beats a frontier model.**

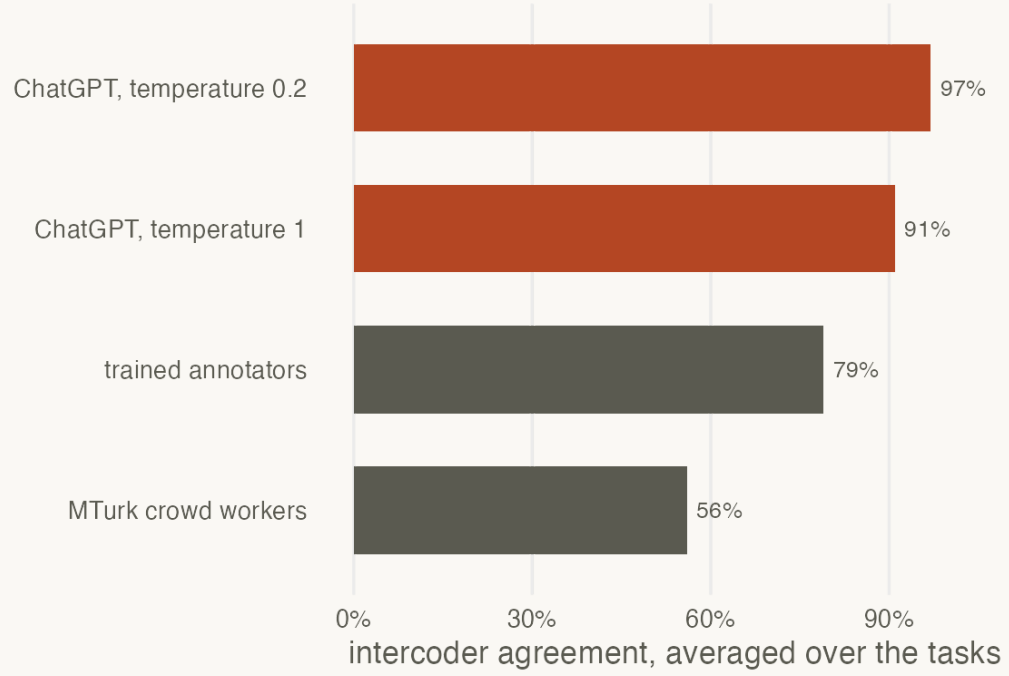
The numbers you report, again

DATA · 300 HELD-OUT ARXIV ABSTRACTS · MODELS IN FREE TEXT

METHOD	ACC.	P CS	R CS	F1 CS	P PHYS	R PHYS	F1 PHYS	P STAT	R STAT	F1 STAT
logistic regression, bag of words · week 3	0.73	**0.63**	**0.67**	**0.65**	0.88	0.91	0.90	**0.67**	**0.60**	**0.63**
logistic regression, sentence embedding · week 3	0.83	0.77	0.75	0.76	0.96	0.98	0.97	0.75	0.75	0.75
Llama 3.2 · 3b, local · zero-shot	0.75	**0.59**	0.93	0.72	0.94	0.99	0.97	0.89	**0.33**	**0.48**
Llama 3.1 · 8b, local · zero-shot	0.76	**0.60**	0.90	0.72	0.95	0.96	0.96	0.88	**0.42**	**0.57**
Gemma 3 · 12b, local · zero-shot	0.74	**0.59**	0.88	0.71	0.92	0.95	0.94	0.83	**0.39**	**0.53**
Claude Haiku 4.5, API · zero-shot	0.76	**0.61**	0.85	0.71	0.96	0.94	0.95	0.79	**0.50**	**0.61**
Claude Opus 4.6, API · zero-shot	0.78	**0.66**	0.80	0.72	0.95	0.95	0.95	0.75	**0.59**	**0.66**

> Values below .70 in colour; for the language models they all sit in two columns, because every model makes the same error: **statistics abstracts are called computer science.** Recall on *statistics* is .33 to .59 for the models, .75 for the classifier that learned arXiv's convention from 4,200 examples.

What the literature says · 1



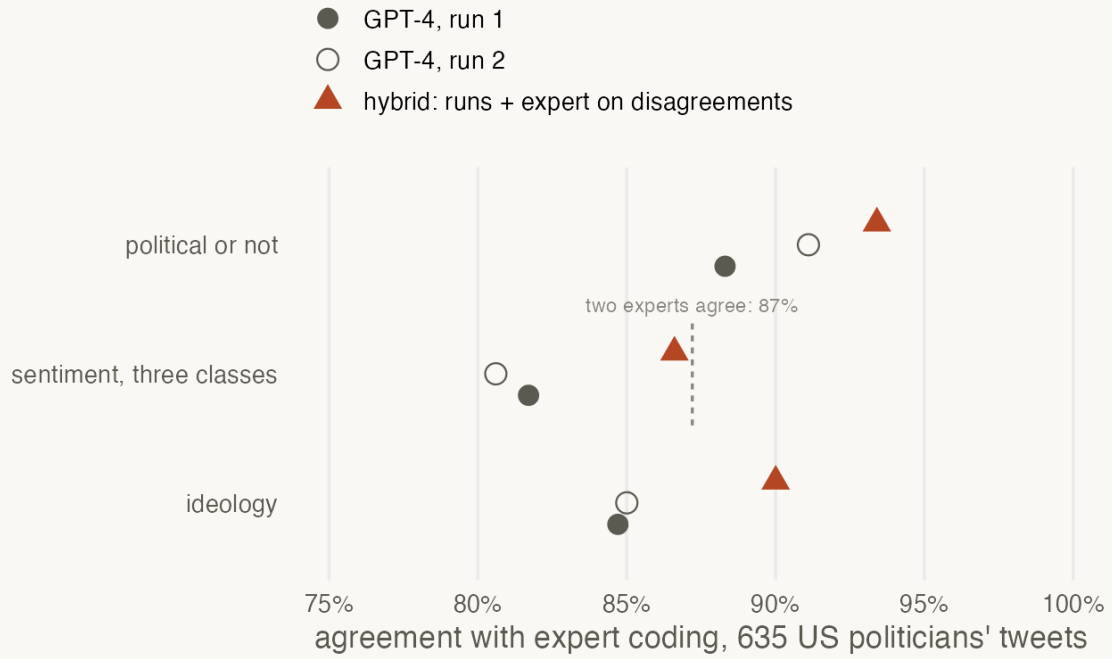
Gilardi et al. (2023), reported averages; the four datasets are 2,382 and 1,856 tweets, 1,606 news articles, and 500 tweets from 2023.

- **Gilardi, Alizadeh & Kubli (2023):** relevance, stance, topic and frame coding of tweets and news. ChatGPT's zero-shot accuracy beat crowd workers by about twenty-five points on average; its agreement with itself beat the trained annotators'; a third of a cent per text
- **Törnberg (2023):** party affiliation from 500 politicians' tweets. GPT-4 beat every individual expert and the crowd majority, with higher reliability — and the same bias toward "Democrat" as the humans
- Both are about GPT versions that no longer exist

> **The early results were real and are not reproducible. Read them for the design: crowd, expert and model on one gold set.**

Gilardi, F., Alizadeh, M., & Kubli, M. (2023). ChatGPT outperforms crowd workers for text-annotation tasks. *PNAS, 120*(30), e2305016120. https://doi.org/10.1073/pnas.2305016120 · Törnberg, P. (2023). ChatGPT-4 outperforms experts and crowd workers in annotating political Twitter messages with zero-shot learning. *arXiv:2304.06588*.

What the literature says · 2



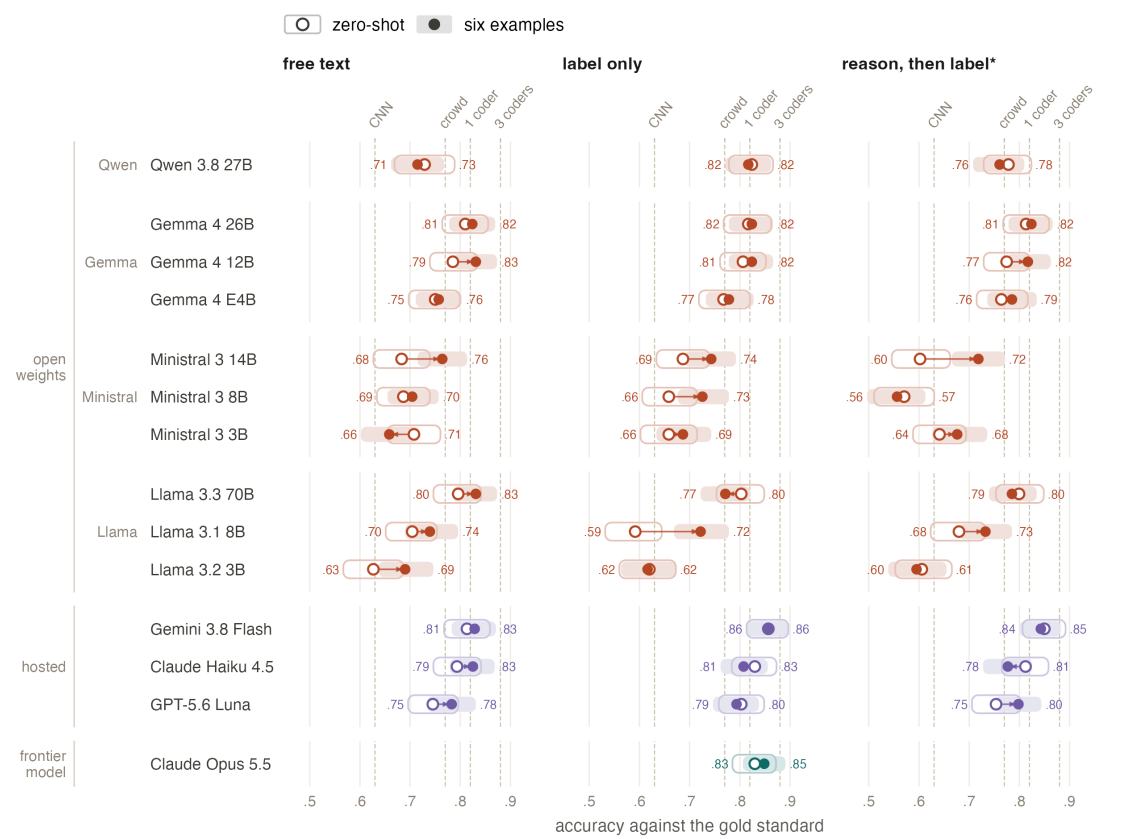
- **Heseltine & Clemm von Hohenberg (2024):** GPT-4 against expert coding of tweets and news in four countries. Binary tasks above ninety percent; three-class sentiment at eighty-one, *below* the two experts' own agreement of eighty-seven
- Their fix is the design to copy: run the model twice, and give only the disagreements — five to thirteen percent of the texts — to a human
- Longer texts and other languages: slightly worse, not dramatically

> **The human baseline decides the verdict. A model at 81 is good when the humans are at 87, and nothing else tells you that.**

Heseltine, M., & Clemm von Hohenberg, B. (2024). Large language models as a substitute for human experts in annotating political text. *Research & Politics, 11*(1). https://doi.org/10.1177/20531680241236239

The row the paper could not add

DATA · 284 DUTCH HEADLINES · FOURTEEN MODELS · MASUR & VAN ATTEVELDT (IN PREP.)



- The 2021 comparison, redone with language models: the same 284 headlines, the same expert gold standard, the paper's codebook in the prompt
- Dashed lines are the 2021 benchmarks: the best classifier, a CNN (.63), crowd coding (.77), one trained coder (.82) and three coders (.88)
- One row per model and one panel per reply format. Open dots without examples, filled dots with six

Masur, P. K., & van Atteveldt, W. (in prep.). Sentiment coding with large language models: Extending the validity comparison of van Atteveldt, van der Velden and Boukes (2021). Figure 1 · van Atteveldt, W., van der Velden, M. A. C. G., & Boukes, M. (2021). The validity of sentiment analysis. *Communication Methods and Measures, 15*(2), 121–140.

What the paper finds

DATA · 284 DUTCH HEADLINES · MASUR & VAN ATTEVELDT (IN PREP.)

- Every model, in its best format, beat every automated method of 2021 without a single coded example. The CNN (.63) was trained on more than 6,000 headlines
- Four models matched one trained coder (.82) without examples. The best, Gemini 3.8 Flash, reached .86; none reached three coders (.88)
- Price bought little: Claude Opus 5.5 cost 25 times as much as Gemini 3.8 Flash and was less accurate. The best open models came within .03
- The reply format moved small models by up to .12 and the strongest by less than .01. Six examples helped weak models and cost some strong ones a little

> **In 2021 the advice was to code by hand or to train a classifier. On this gold standard a prompted model now sits between one trained coder and three.**

Masur, P. K., & van Atteveldt, W. (in prep.). Sentiment coding with large language models: Extending the validity comparison of van Atteveldt, van der Velden and Boukes (2021).

Seconds, hours, and what they cost

MODEL	SECONDS PER SENTENCE	SECONDS PER ABSTRACT	HOURS PER 100,000 ABSTRACTS
Llama 3.2 · 3b, local	0.5	0.6	17.0
Llama 3.1 · 8b, local	0.9	1.7	48.1
Gemma 3 · 12b, local	2.1	3.1	85.4
Claude Haiku 4.5, API	1.2	1.3	37.1
Claude Opus 4.6, API	4.1	3.4	94.7

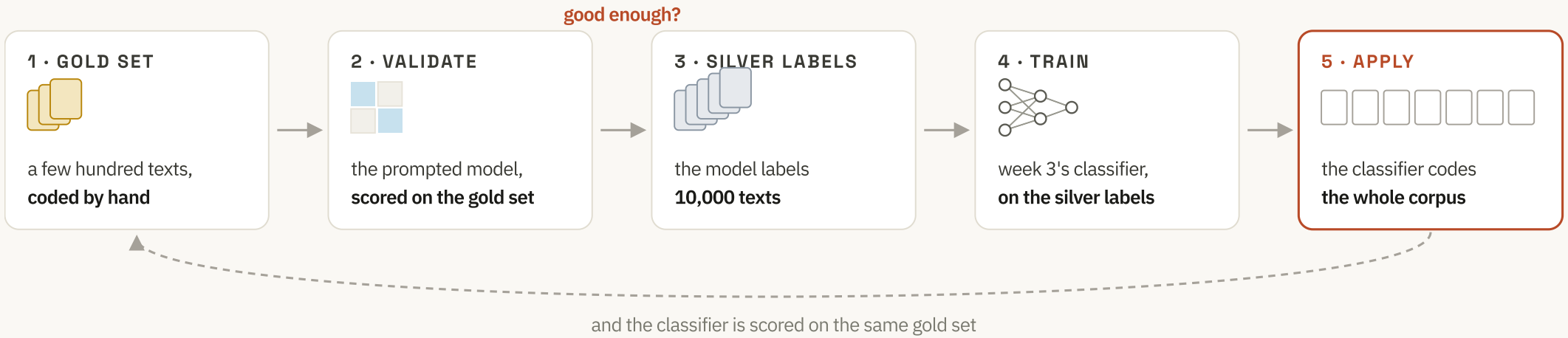
Measured on this laptop while the deck was built; free-text replies, one request at a time.

- A local model costs electricity and time: the largest one here needs two days for a hundred thousand abstracts on a laptop, and hours on a rented GPU
- A frontier model through an API costs cents per thousand tokens — a hundred thousand abstracts is tens to hundreds of euros, and every one of them leaves your machine
- A human coder at two hundred abstracts a day: fifty working weeks
- The Playground's *What the study costs* puts euros, kilowatt-hours and CO_2 on the same page

PLAYGROUND What the study costs

> **Cost is per text and it is real; it is just no longer in coder-weeks. Budget it, and report it with the validation.**

Silver labels



- When the model's score on our gold set is acceptable, its labels are **silver**: not hand-coded, but validated. We use them as the variable itself, or as **training data** for week 3's classifier: cheaper at scale, pinned forever, and often as accurate (Pangakis, Wolken & Fasching, 2023)

> **Either way the hand-coded gold set is what makes the label a measurement. The model does not replace it; it makes it go further.**

Pangakis, N., Wolken, S., & Fasching, N. (2023). Automated annotation with generative AI requires validation. *arXiv:2306.00176*.

VI · What to do with it

Not only text



Climate march, Dam square, Amsterdam, 12 September 2026. Photo: Ainali, Wikimedia Commons, CC BY-SA 4.0; faces pixelated in the original upload.

```
img <- base64_enc(readBin("climate-march.jpg", "raw",
                          file.size("climate-march.jpg")))
prompt <- "Code this photo for a study of protests.
1. Protest: yes or no.
2. Crowd: none, a few people, a group or a crowd.
3. The text on the signs, word for word.
4. The topic of the protest, in three words."

request("http://localhost:11434/api/chat") |>
  req_body_json(list(model = "gemma4:12b", stream = FALSE,
    messages = list(list(role = "user", content = prompt,
                         images = list(img))))) |>
  req_perform() |> resp_body_json() |>
  pluck("message", "content")
```

- The same kind of call, with an image attached: a **vision-language** model reads pixels into the same vector space as words
- The prompt is a codebook again, with four variables for one photo

What the model read

DATA · ONE PHOTO · GEMMA 4, 12 BILLION WEIGHTS, LOCAL

OUTPUT · GEMMA 4, 12 BILLION WEIGHTS, LOCAL

```
1. Protest: yes
2. Crowd: a crowd
3. The text on the signs:
- "KINDEREN VOOR EEN GROENE TOEKOMST"
- "KLIMAAT NU, NIET NA100 JAAR"
- "KLIMAAT NU, NIET NA 100 JAAR"
- "SYSTEM CHANGE"
- "NO MORE"
- "REVOLT"
- "MADAME TISSURE"
- "REVOLT"
4. The topic of the protest: Climate change action
```

- The codes for the photo as a whole are right: a protest, a crowd, and the topic
- The transcription is not. The clearest sign, *Klimaat van rood naar groen doen!*, is missing; *toekomst* completes a banner the frame cuts off; *Madame Tissure* is the waxworks museum's shop sign, misread; and *Revolt*, *No more* and *Klimaat nu, niet na 100 jaar* are nowhere on the photo
- Validation works as it did for text: hand-code a sample of photos, then score each variable on its own

> **The instrument reads images now, and it makes up text the way it makes up rationales. The method is the one we used for text.**

Before the first call

- **Personal data.** Tweets, comments and interviews are personal data under the GDPR. Sending them to a vendor's server is a *transfer*: a processing agreement, a purpose, and usually a data-protection assessment first. Local models avoid the question entirely
- **Terms.** Vendors differ on whether your prompts are kept and whether they train on them. Read the terms of the *API*, not the chat product
- **Disclosure.** The EU AI Act puts duties on providers; on researchers the duty is transparency: which model, which version, which prompt, which date, what it cost — the Instrument block of your AI disclosure
- **Ethics review.** A method that reads and labels people's words at scale is still human-subjects research

> **The first question is not "which model". It is "may this text leave the building". Answer it in writing.**

What the model brings with it

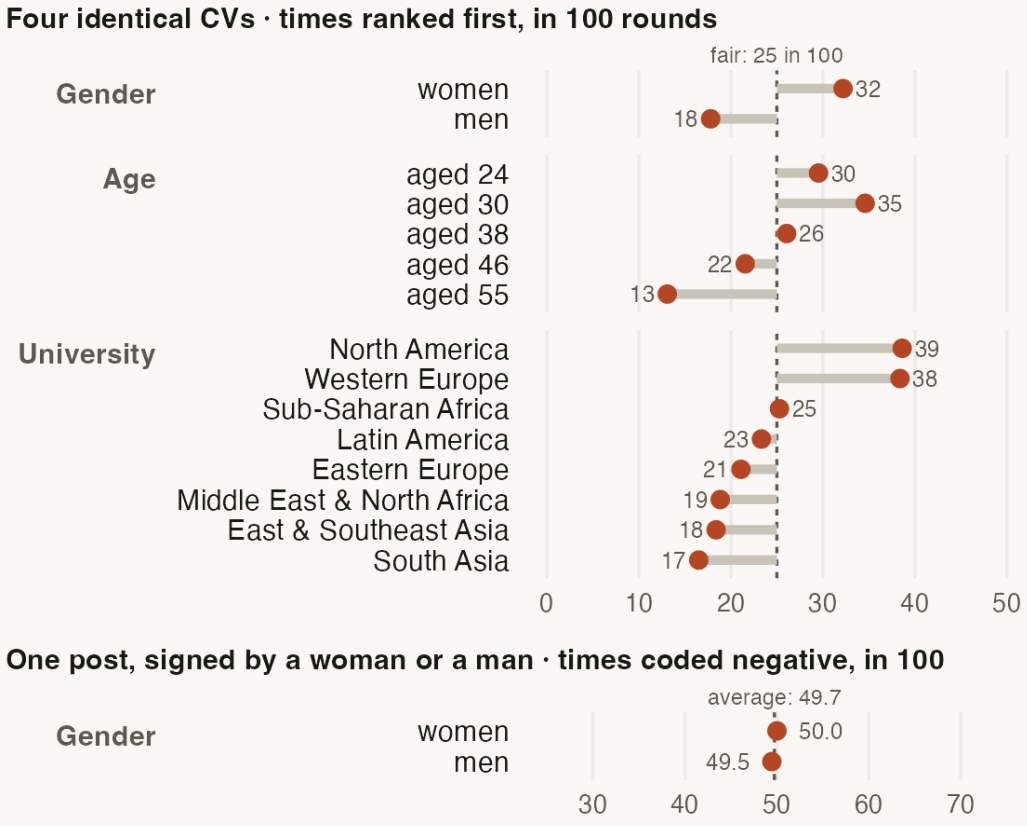
- **Bias.** The model labels the way the text it learned from labels. Törnberg's GPT-4 leaned "Democrat" as the humans did, and a codebook cannot fully correct a prior it cannot see
- **Contamination.** A model that has read the benchmark reproduces its labels: Gilardi et al. lost fourteen points on one task with tweets from *after* the training cutoff
- **Confident nonsense.** The label, the confidence and the rationale are all generated. A fluent reason for a wrong label is the normal case
- **Drift and death.** A closed model is updated in place (Chen et al., 2024) and retired eventually, so any result on it is provisional (Ollion et al., 2024). Open weights stay the same file

> **None of this is an argument against the instrument. It is the list of things the validation has to be designed to catch.**

Chen, L., Zaharia, M., & Zou, J. (2024). How is ChatGPT's behavior changing over time? *Harvard Data Science Review, 6*(2). https://doi.org/10.1162/99608f92.5317da47 · Ollion, É., Shen, R., Macanovic, A., & Chatelain, A. (2024). The dangers of using proprietary LLMs for research. *Nature Machine Intelligence, 6*(1), 4–5. https://doi.org/10.1038/s42256-023-00783-6

Same text, different name

DATA · THE PLAYGROUND'S AUDIT STUDY · LLAMA 3.1 8B



- Four applicants with the same CV, word for word; only name, gender, age, nationality and university differ. A fair ranker puts each first in one round out of four
- Llama 3.1 ranked a woman first 32 times in 100 and a man 18 times; a 30-year-old 35 times, a 55-year-old 13; a North American degree 39 times, a South Asian one 17
- Coding the same post for sentiment, the author's gender moved the share coded *negative* by half a point, and no nationality or name by more than 1.4 points

> **The same model and the same names: a large bias where the text leaves the decision open, almost none where the text decides. Audit the model on the task we use it for.**

What you can now do

- Say what a transformer does with a sentence: tokens, vectors, attention, a stack of layers, one probability per next token
- Explain why a classification by a language model is a next-token prediction, and what that implies for the “confidence” it reports
- Write the loop: a codebook in a prompt, a schema, a map over texts, a confusion matrix against a gold set
- Name what moves a prompt’s score — wording, examples, model, temperature, output format, version — and how each is measured
- Place dictionary, classifier, embeddings and language model on one gold set, and say which you would defend and why

> **Four generations of method scored the same way. That was the plan of the course, and homework 4 does it on your own gold set.**

This week

- **Tuesday · Lab A** — few-shot, structured output and sensitivity: the 48 film reviews under two codebooks, at least two prompts and two models, all on your own laptop
- **Thursday · Lab B** — the validation study: classifier and language model on the paper's headlines; then your week 2 codebook as a prompt on your own gold set
- **Sunday · homework 4** — your own measurement: model vs. dictionary vs. your partner on your 50 tweets, which one you would defend, and a full AI disclosure; started in Lab B
- **Playground** — *The prompt is the instrument* before Tuesday; *Which Harry?* and *What the study costs* whenever
- **Reading** — Gilardi et al. (2023); Heseltine & Clemm von Hohenberg (2024)

> **Everything in the last homework was on a slide today. Install Ollama and pull `llama3.2:3b` before Tuesday.**

Questions?

> If anything is unclear, ask now — or on Tuesday, with a model running on your laptop.

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